# ETAS RTA-BSW v12.7.0

## RTA-J1939 Stack Reference Guide

www.etas.com

## Copyright

The data in this document may not be altered or amended without special notification from ETAS GmbH. ETAS GmbH undertakes no further obligation in relation to this document. The software described in it can only be used if the customer is in possession of a general license agreement or single license. Using and copying is only allowed in concurrence with the specifications stipulated in the contract.

Under no circumstances may any part of this document be copied, reproduced, transmitted, stored in a retrieval system or translated into another language without the express written permission of ETAS GmbH.

**© Copyright 2025** ETAS GmbH, Stuttgart

The names and designations used in this document are trademarks or brands belonging to the respective owners.

ETAS RTA-BSW 12.7.0 - RTA-J1939 Stack Reference Guide R01 EN – 05.2025

# Contents


1 RTA-J1939 . . . . . 5

1.1 Introduction . . . . . 5

2 J1939 Transport Protocol . . . . . 8

2.1 Introduction . . . . . 8

2.2 Supported Features . . . . . 8

2.3 Extensions . . . . . 10

2.4 Limitations . . . . . 14

2.5 Exclusions . . . . . 14

2.6 Deviations . . . . . 14

2.7 API Definition . . . . . 15

2.8 Configuration Advice . . . . . 15

2.9 Integration Advice . . . . . 20

2.10 Hardware Requirements . . . . . 21

3 J1939 Diagnostic Communication Manager . . . . . 22

3.1 Introduction . . . . . 22

3.2 Supported Features . . . . . 25

3.3 Extensions . . . . . 28

3.4 Limitations . . . . . 31

3.5 Exclusions . . . . . 33

3.6 Deviations . . . . . 34

3.7 API Definition . . . . . 34

3.8 Configuration Advice . . . . . 35

3.9 Integration Advice . . . . . 38

3.10 Hardware Requirements . . . . . 39

4 J1939 Request Manager . . . . . 40

4.1 Introduction . . . . . 40

4.2 Supported Features . . . . . 43

4.3 Extensions . . . . . 44

4.4 Limitations . . . . . 46

4.5 Exclusions . . . . . 46

4.6 Deviations . . . . . 46

4.7 API Definition . . . . . 47

4.8 Configuration Advice . . . . . 48

4.9 Integration Advice . . . . . 49
4.10 Hardware Requirements . . . . . 50
5 J1939 Network Management . . . . . 51
5.1 Introduction . . . . . 51
5.2 Supported Features . . . . . 54
5.3 Extensions . . . . . 54
5.4 Limitations . . . . . 57
5.5 Exclusions . . . . . 58
5.6 Deviations . . . . . 58
5.7 API Definition . . . . . 59
5.8 Configuration Advice . . . . . 59
5.9 Integration Advice . . . . . 61
5.10 Hardware Requirements . . . . . 61
6 Glossary . . . . . 62

# 1 RTA-J1939

## 1.1 Introduction

The Society of Automotive Engineers (SAE) J1939 describes a high-speed communications network that allows messages to be exchanged between ECUs. It uses the CAN protocol, which permits any ECU to transmit a message on the network when the bus is idle. Every message includes an identifier that defines the message priority, who sent it, and what data is contained within it.

The sections below provides a brief summary of the key elements of SAE J1939 and introduces the modules that are included in the RTA-J1939 stack.

### 1.1.1 Parameter Groups

*Parameter Groups* (PGs) are a collection of parameters that are conveyed in a J1939 message, and they include commands, data, requests, acknowledgements (ACKs) and negative acknowledgements (NACKs).

SAE J1939 defines a set of PGs, each containing signals with defined content and semantics. The following information is provided for each PG:

- Payload length type: maximal number of bytes, fixed or variable size.
- *Parameter Group Number* (PGN): uniquely identifies a particular Parameter Group and contains the following fields:
    - *Data Page information* (Data Page and Extended Data Page)
    - *8 bit PDU-Format* (PF), identifies one of two PDU Formats that will be used for the messages.
    - *8 bit PDU-Specific* (PS).

PGNs with a PF < 240 (*PDU1* Format) identify “point-to-point” or “destination-specific” messages which must be directed to one or another specific destination, but not to both (e.g. torque control message sent to either the engine or retarder).

PGNs with a PF >= 240 (*PDU2* Format) identify broadcast messages, which can be sent from one or more sources to one or more destinations. The PS field is only relevant for broadcast messages (PDU2 Format), and it is always zero for point-to-point messages (PDU1 Format).

### 1.1.2 CAN Identifiers

RTA-J1939 uses 29-bit CAN identifiers to identify each message. The CAN identifier contains:

- 3-bit Priority
- Parameter Group Number (PGN)
- *Source Address* (SA), contains the address of the ECU that is sending the message.
- *Destination Address* (DA), contains the address of the ECU intended to receive the message (only used for point-to-point messages, not broadcast

messages).

### 1.1.3 Node

Each ECU usually has just one node address, which is used as its Destination Address in incoming received messages, and as the Source Address in messages that it transmits.

However, a single ECU can also implement several different J1939 nodes at once, each with its own node address, and the ECU will accept any of these node addresses as the incoming Destination Address, and can send messages with any of its defined node addresses as the Source Address.

### 1.1.4 NAME

An ECU's *NAME* uniquely identifies that ECU on the network and usually describes the primary vehicle function(s) that it performs.

### 1.1.5 Address Claim

*Address Claim* is the process for claiming a network address and associating an ECU (via its NAME) to that address. The Address Claim process is used by the ECU after power up to claim a network address, and each ECU is capable of announcng which address(es) it intends to use. ECUs can also send requests for Address Claimed to determine the addresses claimed by other ECUs and, thus, determine which ECUs are currently powered up and on the network.

### 1.1.6 RTA-J1939 Modules

The RTA-J1939 stack consists of the following modules, each of which is described in its own sub-section in this Reference Guide:

**J1939 Transport Layer Module (J1939TP)**

- Responsible for the segmentation and reassembly of J1939 PGNs.

**J1939 Request Manager (J1939Rm)**

- Responsible for routing incoming requests to the correct destination, and providing the infrastructure for sending Acknowledgement Message (ACKM) and Request (RQST) PGs.

**J1939 Network Management module (J1939Nm)**

- Responsible for assigning a unique address to each ECU. It also supports the transmission and reception of AddressClaimed PGs and the handling of Address Claimed requests.

**J1939 Diagnostic Communication Manager (J1939Dcm)**

- Responsible for processing diagnostic request messages received via J1939Rm/PduR and sending the appropriate response Acknowledgement Message (ACKM) PGs.

### 1.1.7 RTA-J1939 Architecture

The diagram below shows the position of these RTA-J1939 modules in the AUTOSAR layered architecture.

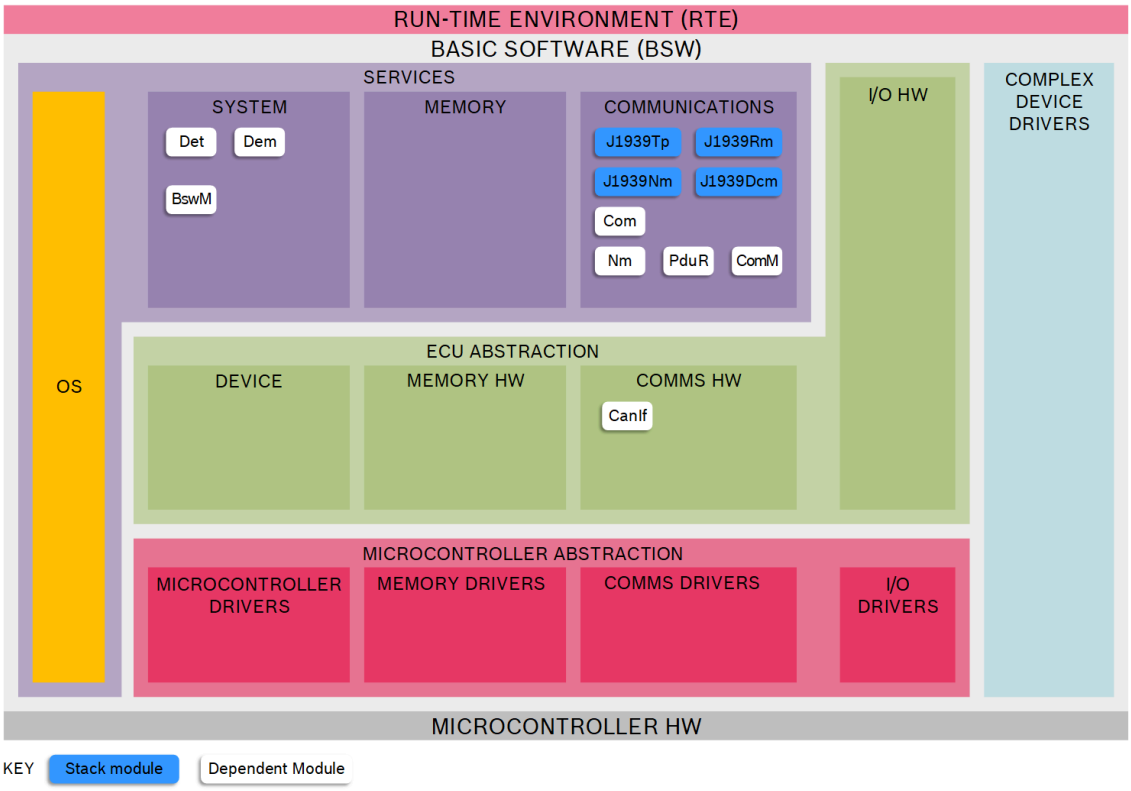


Figure 1.1.RTA-J1939 Architecture

# 2 J1939 Transport Protocol

| | |
|---|---|
| Module name | J1939Tp |
| Package | RTA-J1939 |
| AUTOSAR Module ID | 37 |
| AUTOSAR Version | 22-11 |

## 2.1 Introduction

The J1939 Transport Layer (J1939Tp) is part of the Communication Stack and it deals with the segmentation and reassembly of *J1939 PGNs*.

To assure unique PDU router handling of all J1939 *PGs*, which are independent of the *Payload Length* type (variable or fixed), J1939Tp is used for the transmission of all PGs that are longer than 8 Bytes, regardless of whether their length is fixed or variable.

PGs with variable length that have a configured maximum size of more than 8 bytes, but which do not exceed 8 bytes at runtime, are still transported using J1939Tp, even though no segmentation is necessary and a direct transmission from PDU router to CAN Interface would be possible.

### 2.1.1 Interaction/Dependencies

The diagrams above show the position of J1939Tp (and the other J1939 modules) in the AUTOSAR layered architecture.

J1939Tp has interactions with various other modules, including CanIf and PduR.

**Module Configuration**

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

## 2.2 Supported Features

- Segmentation (via BAM/CMDT) and direct transmission (via PG) of data in transmit direction.
- Reassembly (via BAM/CMDT) and direct reception (via PG) of data in receive direction.
- Control of data flow.
- Timeout supervision.
- Detection of errors during segmentation or reassembly.
- Transmit cancellation.
- Receive cancellation.
- Variable Metadata Length for SduIds and PduIds.
- Concurrent Connection handling.

### 2.2.1 Calibration

As described in the sub-sections below, optional calibration of Timeouts, PGNs and Channels is available through the J1939TpRbCalibrateTimeouts, J1939TpRb-CalibratePgns and J1939TpRbCalibratePgnToSwitchChannel extension parameters.

#### 2.2.1.1 Timeout Calibration Parameters

Calibration (in secs) can be done for the following parameters:

- J1939TpRbTxBamTrMin
- J1939TpRbTxCmdtTr
- J1939TpRbRxBamT1
- J1939TpRbRxCmdtT1.

If calibration is enabled (J1939TpRbCalibrateTimeouts is TRUE), the J1939Tp module, with all the necessary parameters, will be forwarded to the rba_ArxmlGen container by J1939Tp scripts (if rba_ArxmlGen is available). Calibration timeouts are forwarded using the naming convention J1939Tp_Rb[TimeoutType]_PGN.shortname_C, where TimeoutType is TxBamTrMin, TxCmdtTr-Max, RxBamT1Max or RxCmdtT1Max.

#### 2.2.1.2 Priority Calibration Parameter

Calibration can be done for the following parameter:

- J1939TpRbPriority

If calibration is enabled (J1939TpRbCalibratePriority is TRUE), when enabled J1939Tp component with all the necessary parameters will be forwarded to rba_ArxmlGen container by J1939Tp scripts (if rba_ArxmlGen is available). Calibration timeouts are forwarded using the naming convention J1939Tp_RbPriority_PGN.shortname_C, where the calibration is done as integer from 0-7.

#### 2.2.1.3 PGN Calibration Parameters

Calibration can be done for the following parameters:

- J1939TpTxPgPGN
- J1939TpRxPgPGN.

If calibration is enabled (J1939TpRbCalibratePgns is TRUE), the J1939Tp module, with all the RxPgn and TxPgn parameters, will be forwarded to the rba_ArxmlGen container by J1939Tp scripts (if rba_ArxmlGen is available). Calibration TxPgns are forwarded using the naming convention J1939Tp_TxPgPGN_[PGN.shortname]_C, and Calibration RxPgns are forwarded as J1939Tp_RxPgPGN_[PGN.shortname]_C.

DM PGNs are not subjected to this calibration because of their static dependency in J1939Dcm (calibrating a DM PGN in Tp would not provide any benefit as it

cannot be realized in J1939Dcm), so this feature will not forward any DM PGN to rba_arxmlgen. Calibrating any PGN value to a DM PGN is also not advised because it may result in unexpected behaviour in the Upper Layer.

#### 2.2.1.4 Channel Calibration Parameters

Calibration can be done for the following parameters linked to the respective PGs, such that a PG can be switched to another Can channel dynamically:

- J1939TpRxChannel
- J1939TpTxChannel

Calibration of a Channel linked to a PG can be enabled by setting J1939TpRbCalibratePgnToSwitchChannel to TRUE, which enables both Rx and Tx Channel calibration, if at least 2 Rx/Tx Channels are configured.

If calibration is enabled, the J1939Tp component with all the RxChannel and TxChannel parameters will be forwarded to the rba_ArxmlGen container by J1939Tp scripts (if rba_ArxmlGen is available). Calibration of RxChannelId are forwarded using the naming convention J1939Tp_RxChannel_RxPg_[RxPG.shortName]_C, and Calibration TxChannelId are forwarded as J1939Tp_TxChannel_TxPg_[TxPG.shortName]_C.

DM PGNs are not subjected to this calibration because of their static dependency in J1939Dcm (calibrating a DM PGN in Tp would not provide any benefit as it cannot be realized in J1939Dcm), so you should not calibrate the ChannelId for any DM PGN.

## 2.3 Extensions

### 2.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

#### 2.3.1.1 J1939Tp/J1939TpConfiguration/J1939TpRxChannel/J1939TpRxPg/J1939TpRbRxBamT1

| | |
|---|---|
| Short Name | J1939TpRbRxBamT1 |
| Description | This parameter states the timeout in seconds between reception of either two consecutive TP.DT frames or TP.CM_BAM and TP.DT frame on the receiver side for BAM transport protocol variant. |
| Multiplicity | 0..1 |
| Value Range | 0.05..0.75 |
| Default Value | 0.75 |
| Type | ECUC-FLOAT-PARAM-DEF |

#### 2.3.1.2 J1939Tp/J1939TpConfiguration/J1939TpRxChannel/J1939TpRxPg/J1939TpRbRxCmdtT1

| | |
|---|---|
| Short Name | J1939TpRbRxCmdtT1 |
| Description | This parameter states the timeout in seconds between reception of two consecutive TP.DT frames on the receiver side for CMDT transport protocol variant. |
| Multiplicity | 0..1 |
| Value Range | 0.01..0.75 |
| Default Value | 0.75 |
| Type | ECUC-FLOAT-PARAM-DEF |

#### 2.3.1.3 J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpTxPg/J1939TpRbPriority

| | |
|---|---|
| Short Name | J1939TpRbPriority |
| Description | This parameter states the priority to be maintained before sending a Tx frame.User has the flexibility to change priority as per OEM needs. If not configured default value of 7(as per ISO) would be considered. |
| Multiplicity | 0..1 |
| Value Range | 0..7 |
| Default Value | 7 |
| Type | ECUC-INTEGER-PARAM-DEF |

#### 2.3.1.4 J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpTxPg/J1939TpRbTxBamTrMin

| | |
|---|---|
| Short Name | J1939TpRbTxBamTrMin |
| Description | This parameter states the Minimum time gap(in seconds) to be maintained before sending a TP.DT frame in BAM transport protocol variant on the sender side. |
| Multiplicity | 0..1 |
| Value Range | 0.05..0.199 |
| Default Value | 0.05 |
| Type | ECUC-FLOAT-PARAM-DEF |

#### 2.3.1.5 J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpTxPg/J1939TpRbTxCmdtTrMax

| | |
|---|---|
| Short Name | J1939TpRbTxCmdtTrMax |

| | |
|---|---|
| Description | This parameter states the timeout in seconds to send a TP.DT frame on the sender side for CMDT transport protocol variant. |
| Multiplicity | 0..1 |
| Value Range | 0..0.2 |
| Default Value | 0.2 |
| Type | ECUC-FLOAT-PARAM-DEF |

### 2.3.1.6 J1939Tp/J1939TpConfiguration/J1939TpTxChannel/J1939TpTxPg/J1939TpRbTxCmdtTrMin

| | |
|---|---|
| Short Name | J1939TpRbTxCmdtTrMin |
| Description | This parameter states the Minimum time gap(in seconds) to be maintained before sending a TP.DT frame in CMDT transport protocol variant on the sender side. It is advised to keep this value above main function cycle period. |
| Multiplicity | 0..1 |
| Value Range | 0..0.199 |
| Default Value | 0 |
| Type | ECUC-FLOAT-PARAM-DEF |

### 2.3.1.7 J1939Tp/J1939TpGeneral/J1939TpRbCalibrateAddress

| | |
|---|---|
| Short Name | J1939TpRbCalibrateAddress |
| Description | The configuration is used to enable or disable the Address calibration. |
| Multiplicity | 0..1 |
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

### 2.3.1.8 J1939Tp/J1939TpGeneral/J1939TpRbCalibratePgnToSwitchChannel

| | |
|---|---|
| Short Name | J1939TpRbCalibratePgnToSwitchChannel |
| Description | This configuration parameter is used to enable or disable the calibration feature to switch the PGNs configured under a channel to a different channel. |
| Multiplicity | 0..1 |
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

#### 2.3.1.9 J1939Tp/J1939TpGeneral/J1939TpRbCalibratePgns

| | |
|---|---|
| Short Name | J1939TpRbCalibratePgns |
| Description | The configuration is used to enable or disable the Pgn calibration. |
| Multiplicity | 0..1 |
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

#### 2.3.1.10 J1939Tp/J1939TpGeneral/J1939TpRbCalibratePriority

| | |
|---|---|
| Short Name | J1939TpRbCalibratePriority |
| Description | The configuration is used to enable or disable the priority calibration parameter. |
| Multiplicity | 0..1 |
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

#### 2.3.1.11 J1939Tp/J1939TpGeneral/J1939TpRbCalibrateTimeouts

| | |
|---|---|
| Short Name | J1939TpRbCalibrateTimeouts |
| Description | The configuration is used to enable or disable the timeout calibration parameters. |
| Multiplicity | 0..1 |
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

#### 2.3.1.12 J1939Tp/J1939TpGeneral/J1939TpRbMaxNumofChannel

| | |
|---|---|
| Short Name | J1939TpRbMaxNumofChannel |
| Description | The parameter is used to define the maximum number of channels used by J1939Tp |
| Multiplicity | 1..1 |
| Value Range | 1..255 |
| Default Value | - |
| Type | ECUC-INTEGER-PARAM-DEF |

#### 2.3.1.13 J1939Tp/J1939TpGeneral/J1939TpRbTxConfTimeout

| | |
|---|---|
| Short Name | J1939TpRbTxConfTimeout |

| | |
|---|---|
| Description | Timeout in seconds for the CanIf Tx confirmation. After this time the J1939Tp assumes that an N-PDU could not be transmitted. Please note: The Tx confirmation timeout should be set to a value that enabled detection of a lost Tx confirmation in time, and that ensures that normal transmission delay caused by lower message priority does not lead to an error. |
| Multiplicity | 1..1 |
| Value Range | 0..65.535 |
| Default Value | - |
| Type | ECUC-FLOAT-PARAM-DEF |

### 2.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

- There are no additional API defined in this version.

## 2.4 Limitations

- Retry is not supported if TxConfirmation is not received.

- The J1939Tp_ChangeParameter API is implemented such that if ChangeParameter is called for a RxSdu, then the CTS contents of only that RxSdu is changed. It cannot be guaranteed, however, that the next reception request will use the same RxSdu.

## 2.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- There are no known exclusions in this version.

## 2.6 Deviations

### 2.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

- There are no known deviations in this version.

## 2.7 API Definition

Table 2.15.J1939Tp Supported API functions

| | |
|---|---|
| `J1939Tp_MainFunction` | Main function of the J1939 Transport Layer used for scheduling purposes and timeout supervision. |
| `J1939Tp_Init` | This function initializes the J1939Tp module. |
| `J1939Tp_Shutdown` | This function is used to shutdown the J1939Tp module. |
| `J1939Tp_GetVersionInfo` | Returns the version information of the J1939Tp module. |
| `J1939Tp_Transmit` | This service function is used to request the transfer of a J1939Tp N-SDU. |
| `J1939Tp_CancelTransmit` | This service function is used to cancel the ongoing transmission of a J1939Tp NSDU. |
| `J1939Tp_CancelReceive` | This service function is used to cancel the ongoing reception of a J1939Tp NSDU. |
| `J1939Tp_ChangeParameter` | This service function is used to change reception parameters of J1939Tp for a specific N-SDU. |
| `J1939Tp_RxIndication` | Indication of a received I-PDU from a lower layer communication interface module. |
| `J1939Tp_TxConfirmation` | The lower layer communication interface module confirms the transmission of an IPDU. |

For further information on these functions, please refer to AUTOSAR_SWS_SAE-J1939TransportLayer.pdf

## 2.8 Configuration Advice

**NOTE**

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the “Configuration Generation” section of the RTA-CAR User Guide.

### 2.8.1 General Advice

- J1939Tp supports configurable timer parameters T1 and Tr for BAM and CMDT connections. These timer parameters can be configured at the Rx/Tx PG level. For backwards compatibility, the configuration of these timers is not mandatory. If they are not configured, they will have the default values mentioned in SAE J1939-21 (2006-12), Data Link Layer.

### 2.8.2 Code Generation based on configuration class

J1939Tp supports 3 kinds of configuration class

- Pre-compile
- Post-build Loadable
- Post-build Selectable

If either of the two Post-Build options is to be used for J1939Tp, the following two configuration steps must first be completed.

**Step 1** Select the Configuration Class

Go to the J1939Tp module configuration, and then the “properties” tab.

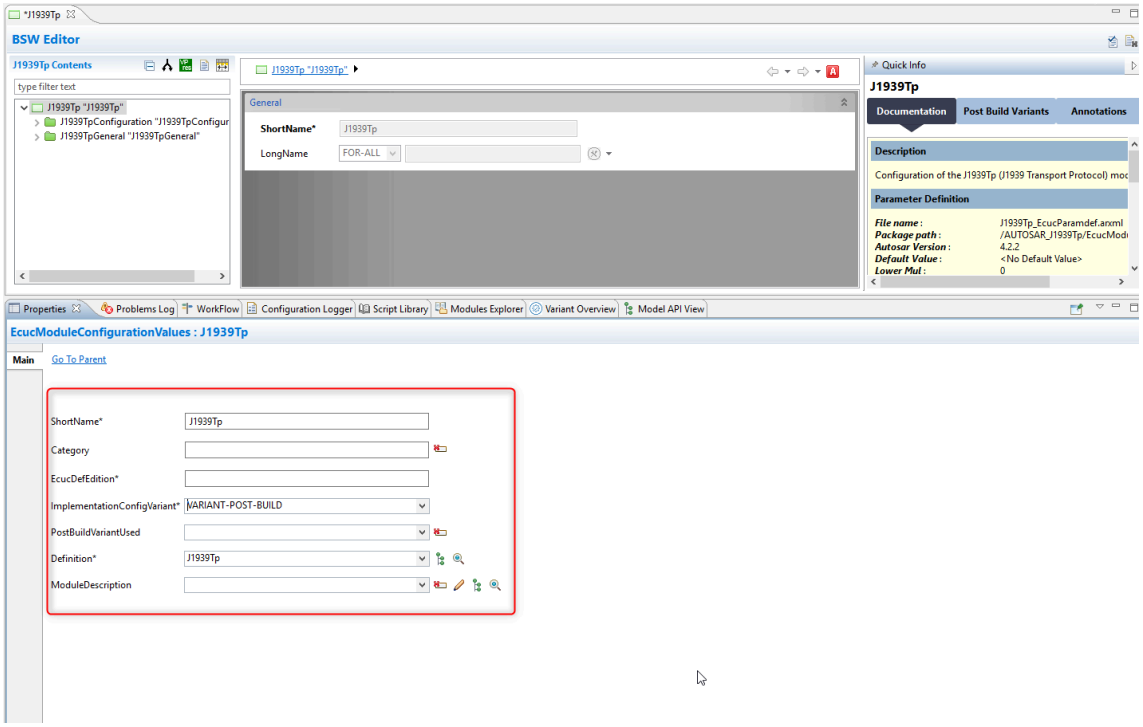


Figure 2.1.Select which configuration class of J1939Tp

ImplementationConfigVariant will be set to “variant-pre-compile” by default, but if either of the two Post-Build options will be used for J1939Tp, you must select the generic “variant-post-build” option (and not the specific “variant-post-build-loadable” or “variant-post-build-selectable” options).

**Step 2** EcuM Configuration

If ImplementationConfigVariant is set to “variant-post-build”, the EcuM configuration for J1939Tp and all its dependent modules must be configured with POST-BUILD_PTR.

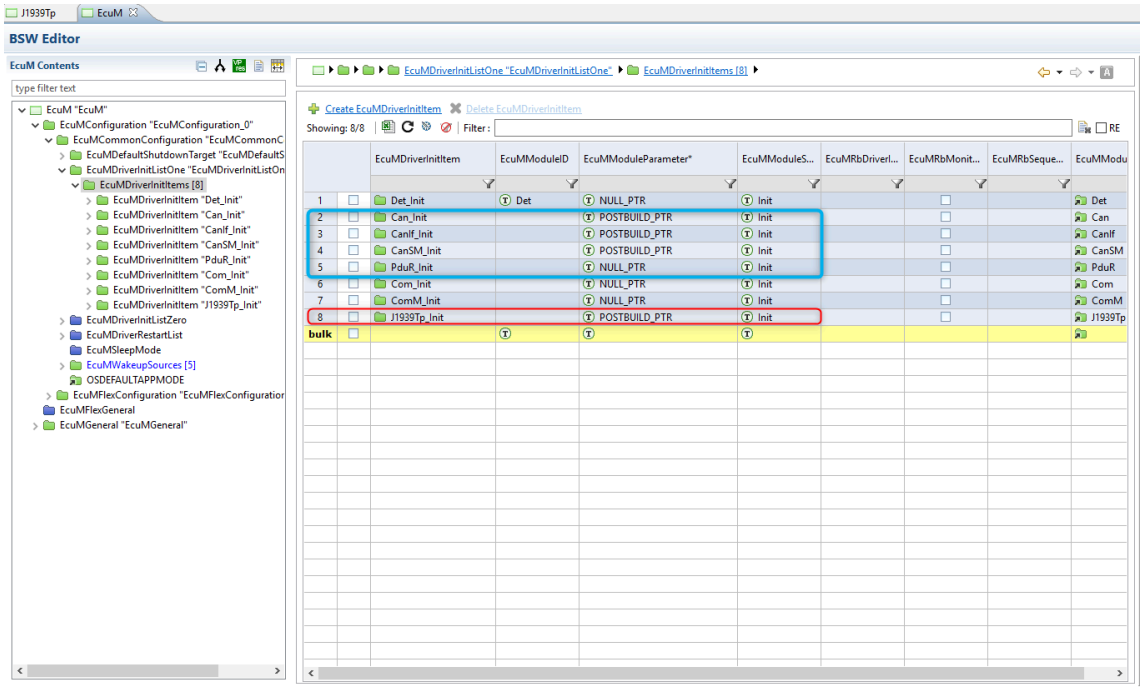


Figure 2.2.EcuM configuration

### 2.8.2.1 Code Generation Options

Depending on the above setting of ImplementationConfigVariant, the following options can be deployed at Code Generation.

**Pre-Compile**

If this option is chosen, all ECUs using the generated BSW will have the same configuration and there is no way to change this configuration without re-generating and re-compiling the whole of the ECU's software.

As shown below, select the "J1939Tp_PreCompile" option for the J1939Tp module. The other options on the right can be set at your discretion.

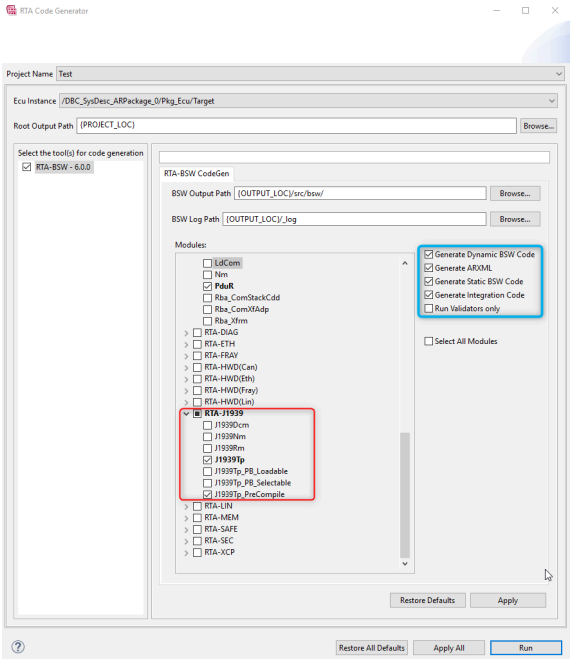


Figure 2.3.Generate J1939Tp with Configuration Class as Pre-compile

### Post-Build Loadable

This option results in the generation of only the configuration for the J1939Tp module, and it does not generate any of the other code.

Post-Build Loadable is only available after the J1939Tp module's main code has initially been fully generated in a Post-Build configurable form, and this is achieved with a one-time Code Generation using Post-Build Selectable, as shown here.

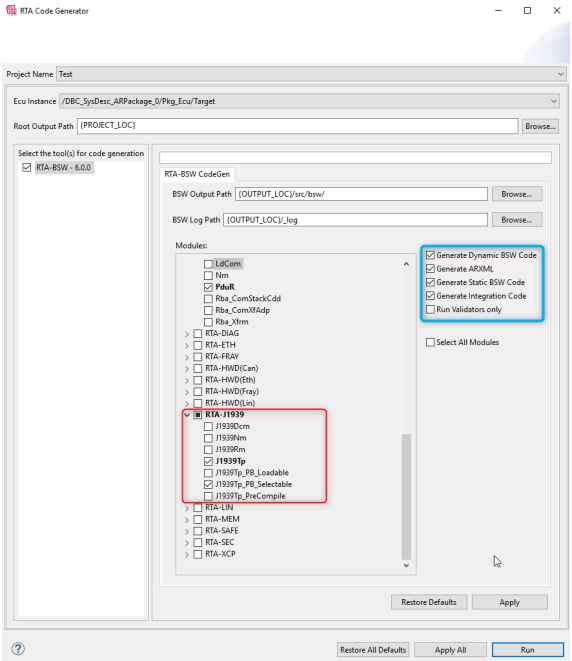


Figure 2.4.Full Code Generation with Post-Build

Then, if the module's configuration is subsequently changed, as shown below the Post-Build Loadable option can be used so that only the configuration files are re-generated (the main J1939Tp code will not be re-generated).

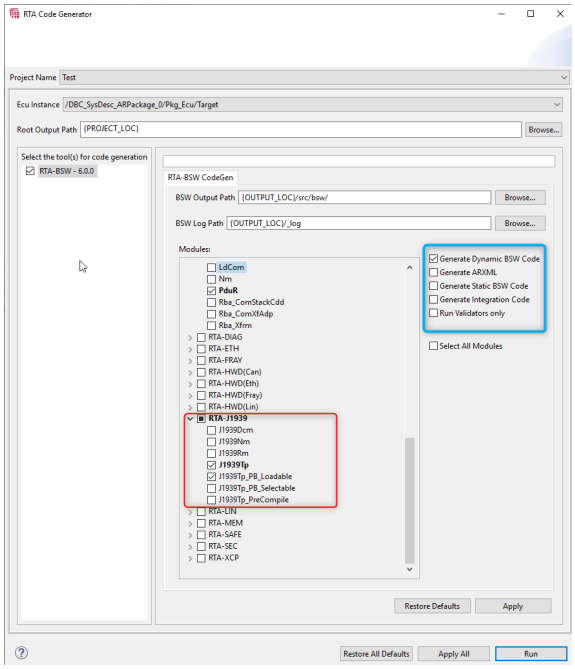


Figure 2.5.Regeneration for Post-Build

Note that the “Delete Existing Output Folder” option must be un-ticked when using Post-Build Loadable. This results in the loadable parts being allocated into a different folder from the full generation that was previously created by the initial Post-Build Selectable.

**Post-Build Selectable**

This option is used when multiple configurations for J1939Tp will be created, and the required configuration can be selected at run time.

As shown below, the J1939Tp_PB_Selectable option is ticked.

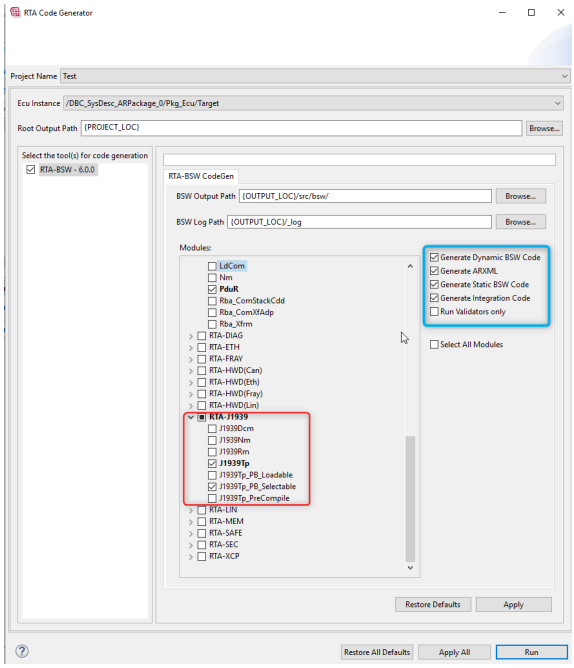


Figure 2.6.Code Generation with Selectable Post-Build

## 2.9 Integration Advice

### 2.9.1 Init Functions

`J1939Tp_Init` is called by EcuM during the system initialization

### 2.9.2 DeInit Functions

`J1939Tp_Shutdown` function is required to de-initialize all the RAM areas of the previously active config set. This function is used only when Post-build is enabled

### 2.9.3 Scheduled Functions

`J1939Tp_MainFunction` is called cyclically for scheduling purposes and timeout supervision.

### 2.9.4 Misc

CAN events (receive and transmit) may be configured in either polling or interrupt mode.

If the CAN configuration mixes polling and interrupt mode, then you should be aware of the CMDT timing parameters when scheduling the cyclic main functions for the CAN stack.

In particular, if transmit confirmation messages are handled by periodic polling, and received messages are handled by interrupts, polling must be frequent enough that each transmit confirmation is handled before any response to the transmitted data can be received. Connections will be aborted with a 'busy' message by J1939Tp if these messages are handled in any other order.

### 2.9.5 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The J1939Tp module uses the following OS resources:

- J1939Tp
- J1939Tp

### 2.9.6 Post Build Selectable

When Post-build Selectable is used with multiple variants, target variant can be selected by using user defined API - EcuM_DeterminePbConfiguration from EcuM integration. Please refer relevant integration advice from ECU State Management for more detail information.

## 2.10 Hardware Requirements

Table 2.16.J1939Tp Hardware Requirements

| Support | Usage |
|---|---|
| FLASH | 11016 bytes |
| RAM | 201 bytes |

Usage calculated based on a sample configuration.

# 3 J1939 Diagnostic Communication Manager

| | |
|---|---|
| Module name | J1939Dcm |
| Package | RTA-J1939 |
| AUTOSAR Module ID | 58 |
| AUTOSAR Version | 23-11 |

## 3.1 Introduction

The J1939 Diagnostic Communication Manager (J1939Dcm) is responsible for processing diagnostic request messages received via J1939Rm/PduR and sending the appropriate response *Acknowledgement Message (ACKM) PGs*.

### 3.1.1 Reception of a Request

The J1939Dcm Main Function invokes the requested Diagnostic Message when a DM service is triggered by any of the following -

- `J1939Dcm_RequestIndication`
- `J1939Dcm_RxIndication`
- `J1939Dcm_DemTriggerOnDTCStatus`
- `J1939Dcm_StartOfReception`
- `J1939Dcm_CopyRxData`
- `J1939Dcm_TpRxIndication`
- The DEM Module

The DM service calls the appropriate DEM APIs to complete its operation, then either sends an acknowledgement to J1939Rm or transmits the response by calling `PduR_Transmit`. After transmission, the buffer is released, the *State Machine* is reset, and this sequence repeats. For *Memory Access Operations*, the State Machine is reset only after the *Memory Access Operation Sequence* is completed.

### 3.1.2 Transmission of a Response

J1939Dcm either sends an acknowledgement to J1939Rm (*Positive, Negative, Cannot Respond*) or it transmits the response by calling `PduR_J1939DcmTransmit`. The response for a Status Request Command for a DM14 message is sent via the `J1939Dcm_StatusRequest` API.

### 3.1.3 Summary Diagrams

The following diagrams summarise the reception of request Diagnostic messages and the transmission of Diagnostic messages:

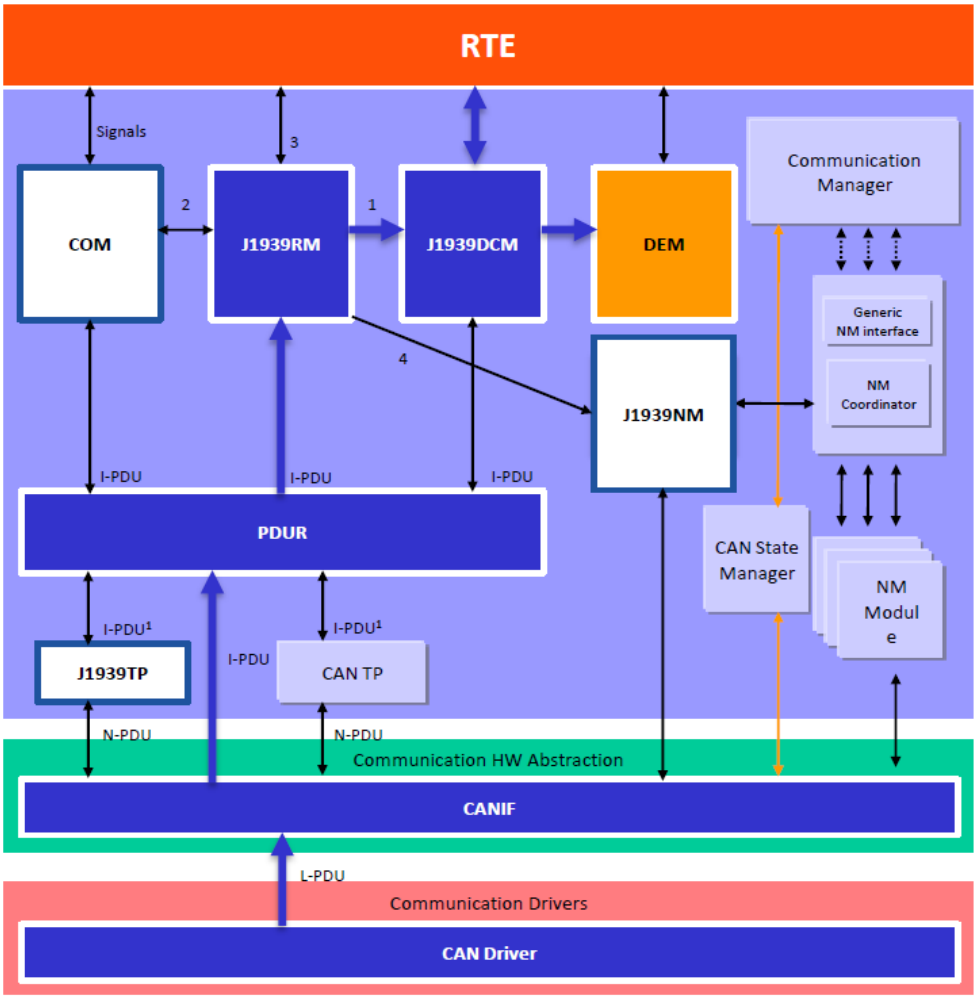


Figure 3.1.Reception of request Diagnostic message

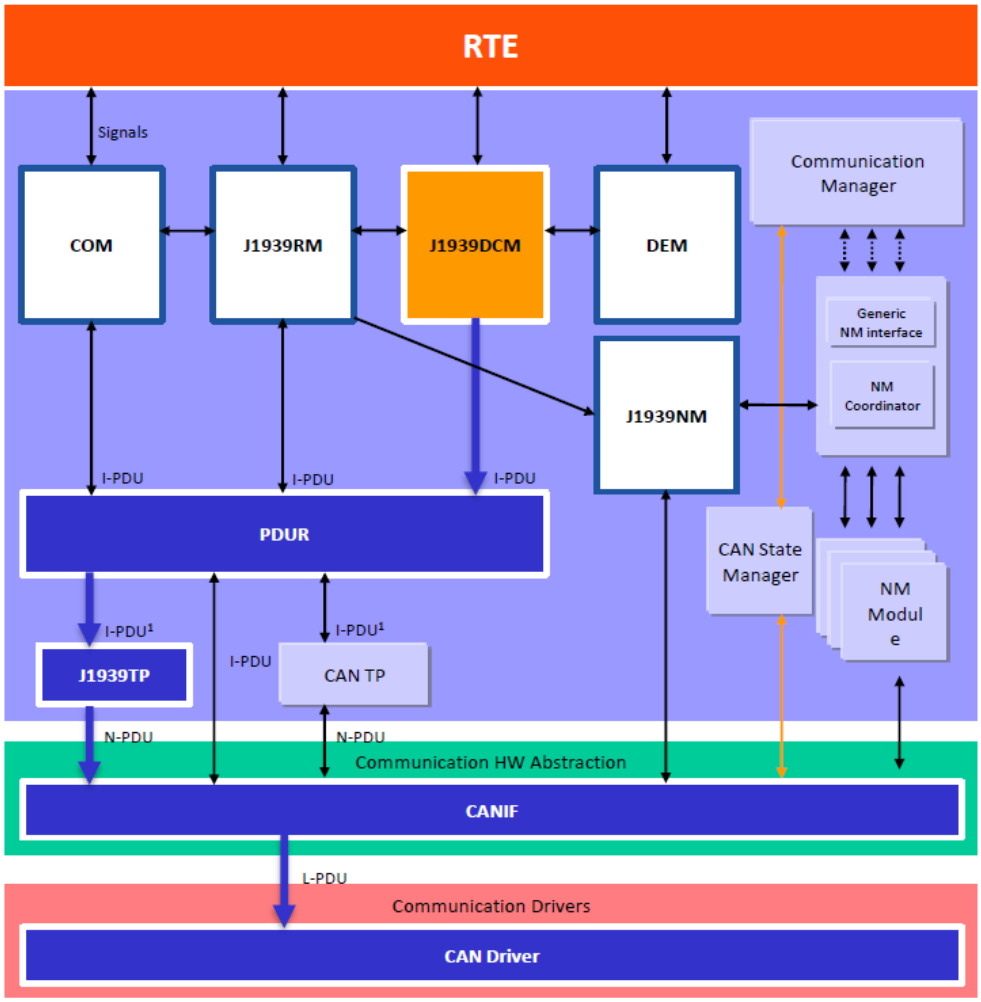


Figure 3.2.Transmission of Diagnostic message

### 3.1.4 Interaction/Dependencies

The diagrams above show the position of J1939Dcm (and the other J1939 modules) in the AUTOSAR layered architecture. The interaction/dependencies of J1939Dcm with other modules is summarised here:

**J1939Rm**

- The J1939Dcm API `J1939Dcm_RequestIndication` is triggered by J1939Rm.
- J1939Dcm calls the `J1939Rm_SendAck` API to send any acknowledgements.

**PduR**

- J1939 calls the PduR API `PduR_J1939DcmTransmit` for the transmission of response bytes.
- PduR calls `J1939Dcm_CopyTxData` to get the required data for transmission and then calls either `J1939Dcm_Txconfirmation` or

`J1939Dcm_TpTxConfirmation` to indicate the result of the transmission.

**BswM**

- If it is configured, the DM13 service uses the API `BswM_J1939DcmBroadcastStatus` to alter the Broadcasting functionality.

**Other**

- J1939Dcm also has interactions with the J1939Nm, Det, Dem, SchM, ComM and CanIf modules.

**Module Configuration**

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

## 3.2 Supported Features

A typical user of J1939Dcm will use one or more DM services. J1939Dcm currently supports the following standard SAE J1939 Diagnostic Messages (DMs).

- DM01
- DM02
- DM03
- DM04
- DM05
- DM06
- DM11
- DM12
- DM13
- DM14
- DM15
- DM16
- DM18
- DM19
- DM20
- DM21
- DM23
- DM24
- DM25
- DM26
- DM29
- DM31

- DM32
- DM53
- DM54

### 3.2.1 Additional Supported Messages

In addition to the standard messages listed above, J1939Dcm also supports the following inter-dependent services used for Memory Access Operations.

**DM14**

This service is triggered from the Core State Machine after the request is received via the PduR through the `J1939Dcm_StartOfReception`, `J1939Dcm_CopyRxData` and `J1939Dcm_TpRxIndication` APIs. It is used to send the request message for Memory Access Operation.

It is also used to send the Key for the respective Seed if security is required (the key would be then validated and the response is sent using the DM15 service).

DM14 doesn't transmit any response data bytes. This message is always followed by a DM15 service response message (except in the case of Operation complete).

**DM15**

This service is triggered by the Memory Access State Machine to send the Response for a Memory Access Operation. It is used to send the response for the command received via Memory Access Request or to send the Seed if security is required.

DM15 is also used to close the Sequence after the Memory Access Operation is completed or failed. This service doesn't receive any request data bytes. Response data bytes are transmitted via the PduR.

In DM14 request, the SAE reserved bit is sent as 1, so in the DM15 response, the reserved bits (bit 1 and 5 of byte2) are set to 1.

**DM16**.

This service is triggered from the Core State Machine after the request is received via the PduR through the `J1939Dcm_StartOfReception`, `J1939Dcm_CopyRxData` and `J1939Dcm_TpRxIndication` APIs. It is used to perform the Binary Data Transfer. Response data bytes are transmitted via the PduR. The response message length may be greater than 8 bytes, in which case J1939Tp will be used.

In the case of Spatial Addressing, the number of Occurrences of Raw Binary Data in the DM16 response (read operation) will specify the actual length of Number of Objects.

**DM18**

This service is triggered after the request containing the Long Key or Session Key is received via the PduR through the `J1939Dcm_StartOfReception`, `J1939Dcm_CopyRxData` and `J1939Dcm_TpRxIndication` APIs. It is triggered by the Memory Access State Machine to send the Long Seed/Certificate if Long Seed Security is required (it is only used if Long Seed Security is required). Response

data bytes are transmitted via the PduR. The response message length may be greater than 8 bytes, in which case J1939Tp will be used.

**DM23**

This service communicates information related to previously MIL-On emission-related diagnostic trouble codes preceded by the diagnostic lamp status.

**DM32**

The DM32 provides the DTCs and associated timers related to a regulated exhaust emission level exceedance due to an emission control system malfunction. The DTCs that are reported may be active or previously active. The J1939Dcm will call the Appl functionJ1939DcmAppl_GetDM32DTCsandTimers to fetch the relevant data.

**DM53**

The DM53 provides information that is limited to the active service only diagnostic trouble codes which do not display an operator indication lamp and are intended for use at the service location. It is used to notify the service technician about information that may be helpful when troubleshooting the reported condition. These DTCs are not designed for vehicle operator indication display when the vehicle is in normal use, but are intended for when the vehicle is being repaired. The data contains a list of diagnostic codes and occurrence counts for the active trouble codes.

**DM54**

The DM54 provides information that is limited to the previously active service only diagnostic trouble codes and that are intended only for use at the service location for the product. It is used to notify the service technician about conditions that may be helpful when troubleshooting the reported condition. These DTCs are not designed for vehicle operator indication display when the vehicle is in normal, but are intended for when the vehicle is being repaired. The data contains a list of diagnostic codes and occurrence counts for previously active trouble codes.

### 3.2.2 Stopping DM01 Periodic Message

The `J1939DcmRbAllowApplToStopDM1PeriodicMsg` extension configuration parameter can be used to enable/disable the DM01 Periodic Message feature. This parameter is optional and should be configured only if DM01 is supported within the project. See Configuration Advice Section 3.8 for more details.

### 3.2.3 Support for DM01 Broadcasting on Multiple Nodes

This feature allows the switching of nodes and transmission of a DM01 response to either periodic messages or DTC status changes via the configuration parameter `J1939DcmRbDM01MultiNodeResponse`. See Configuration Advice Section 3.8 for more details.

### 3.2.4 Disabling the acceptance/rejection of new DM request Messages

If the application has no requirement to accept or reject new DM request messages on certain conditions, the `J1939DcmRbGetPermToAcceptReq` configura-

tionextension parameter can be used to disable this functionality. This parameter is optional and should be configured only if the application wants to accept or reject new DM request Messages. See the Configuration Advice section for more information.

### 3.2.5 Request Queuing

To serialize the use of Dem, you can optionally enable the queuing of Periodic/EventTriggered DM01 requests and DMx requests for DM02/DM03/DM06/DM12/DM23.

- When a DM01 periodic transmission is active, and a new DMx request is received for either DM02/DM03/DM06/DM12/DM23, the DMx request should be queued instead of sending NACK via J1939Rm.
- When either of DM02/DM03/DM06/DM12/DM23 is active, and a new request is received for DM01 via periodic scheduler or Event triggered via Dem, the DM01 request should be queued instead of ignoring the request.
- When a Tester Triggered DM01 request is active, and the Periodic Transmission of DM01 is due for that cycle, the Periodic transmission request should be queued instead of being ignored.
- When a Tester Triggered DM01 request is active, and a new request is received for DM01 via J1939Dcm_DemTriggerOnDTCStatus, the Dem Triggered request should be queued instead of being ignored.
- When a Periodic Transmission of DM01 is queued, and a new request is received for DM01 via J1939Dcm_DemTriggerOnDTCStatus, the Dem Triggered request should be prioritized for queuing.

### 3.2.6 Checking Broadcast Status

The J1939Dcm_DM13_ReadBroadcastStatusInfo extension API is provided to allow you to check whether or not broadcasting is stopped. A return of 0xFFFF indicates that broadcasting is allowed, any other return indicates that it is not.

### 3.2.7 Generic DMx Support

This feature supports projects that implement Diagnostic Messages in the Application Software itself. It is enabled by setting `J1939DcmGenericDMxSupport` to TRUE. Please see the Integration Advice section, as well as the Configurable Interfaces section in the AUTOSAR v22-11 documententation, for more details.

## 3.3 Extensions

### 3.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

#### 3.3.1.1 J1939Dcm/J1939DcmGeneral/J1939DcmRbAllowApplToStopDM1PeriodicMsg

| | |
|---|---|
| Short Name | J1939DcmRbAllowApplToStopDM1PeriodicMsg |
| Description | If enabled, this feature provides the Application an API to allow/stop DM01 Periodic messages for the current cycle. Application API 'J1939DcmAppl_AllowDM1PeriodicMsg' would be invoked from J1939Dcm_DM1Scheduler and J1939Dcm_DemTriggerOnDTCStatus. The Application can return E_OK to allow the transmission of DM01 Periodic messages, or return E_NOT_OK if DM01 periodic transmission is to be stopped for the current cycle. |
| Introduction | * True : Enabled (ON); * False : Disabled (OFF) |
| Multiplicity | 0..1 |
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

#### 3.3.1.2 J1939Dcm/J1939DcmGeneral/J1939DcmRbDM01MultiNodeResponse

| | |
|---|---|
| Short Name | J1939DcmRbDM01MultiNodeResponse |
| Description | Broadcast DM01 on any node independent of the configuration. |
| Multiplicity | 0..1 |
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

#### 3.3.1.3 J1939Dcm/J1939DcmGeneral/J1939DcmRbGetPermToAcceptReq

| | |
|---|---|
| Short Name | J1939DcmRbGetPermToAcceptReq |
| Description | If enabled, this feature provides the Application an API to accept/reject DM Request messages received. Application API 'J1939DcmAppl_GetPermToAcceptReq' would be invoked from J1939Dcm_StartOfReception, J1939Dcm_RequestIndication and J1939Dcm_RxIndication APIs. The Application can return E_OK to accept the request message, or return E_NOT_OK if the request is to be rejected. If this feature is enabled, an Application callback 'J1939DcmAppl_RequestCompleted' will be given after the TxConfirmation to inform the Application that the request was serviced completely |
| Multiplicity | 0..1 |

| | |
|---|---|
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

#### 3.3.1.4 J1939Dcm/J1939DcmGeneral/J1939DcmRbMemMaxNumberOfRetryForTransmit

| | |
|---|---|
| Short Name | J1939DcmRbMemMaxNumberOfRetryForTransmit |
| Description | Maximum number of times the Device should retry to Transmit the Response for Memory Access Services when Status Request is being Transmitted and vice-versa.( configure this value as multiple of J1939Dcm Main Function Period) |
| Multiplicity | 0..1 |
| Value Range | 0..255 |
| Default Value | - |
| Type | ECUC-INTEGER-PARAM-DEF |

#### 3.3.1.5 J1939Dcm/J1939DcmGeneral/J1939DcmRbMemTimeOutForData

| | |
|---|---|
| Short Name | J1939DcmRbMemTimeOutForData |
| Description | Time-out period while waiting for data for requested operation(write).( configure this value as multiple of J1939Dcm Main Function Period) |
| Multiplicity | 0..1 |
| Value Range | 0..1 |
| Default Value | - |
| Type | ECUC-FLOAT-PARAM-DEF |

#### 3.3.1.6 J1939Dcm/J1939DcmGeneral/J1939DcmRbMemTimeOutForOperationComplete

| | |
|---|---|
| Short Name | J1939DcmRbMemTimeOutForOperationComplete |
| Description | Time-out period while waiting to receive Operation Complete Message from Tool.( configure this value as multiple of J1939Dcm Main Function Period) |
| Multiplicity | 0..1 |
| Value Range | 0..1 |
| Default Value | - |
| Type | ECUC-FLOAT-PARAM-DEF |

#### 3.3.1.7 J1939Dcm/J1939DcmGeneral/J1939DcmRbMemTimeOutForOperationExecution

| | |
|---|---|
| Short Name | J1939DcmRbMemTimeOutForOperationExecution |

| | |
|---|---|
| Description | Time-out period while waiting for requested operation(read, write, erase) to be completed from application side.( configure this value as multiple of J1939Dcm Main Function Period) |
| Multiplicity | 0..1 |
| Value Range | 0..1 |
| Default Value | - |
| Type | ECUC-FLOAT-PARAM-DEF |

#### 3.3.1.8 J1939Dcm/J1939DcmGeneral/J1939DcmRbMemTimeOutFor-ReceivingKey

| | |
|---|---|
| Short Name | J1939DcmRbMemTimeOutForReceivingKey |
| Description | Time-out period while waiting to receive Key from Tool.( configure this value as multiple of J1939Dcm Main Function Period) |
| Multiplicity | 0..1 |
| Value Range | 0..1 |
| Default Value | - |
| Type | ECUC-FLOAT-PARAM-DEF |

### 3.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

#### 3.3.2.1 J1939Dcm_DM13_ReadBroadcastStatusInfo

| | |
|---|---|
| Syntax | `J1939Dcm_DM13_ReadBroadcastStatusInfo(void)` |
| Retvals | 0xFFFF - Broadcasting is allowed.<br>Anything other than 0xFFFF - Broadcasting is stopped. |
| Description | Returns information about the current broadcasting status info. |

## 3.4 Limitations

Currently 23 DM services are implemented in J1939Dcm: DM01, DM02, DM03, DM04, DM05, DM06, DM11, DM12, DM13, DM14, DM15, DM16, DM18, DM19, DM20, DM21, DM23, DM24, DM25, DM26, DM29, DM31 and DM32.

### 3.4.1 Parallel Processing

Parallel processing is supported in J1939Dcm with the DM01 service, with the following considerations.

When DM01 is running in the ECU, parallel processing is only allowed for the following DM services: DM04, DM05, DM11, DM13, DM14, DM15, DM16, DM18, DM19, DM20, DM21, DM24, DM25 and DM26.

The DM01 service cannot be active in parallel with the following services for the reasons stated -

- DM02, because they use the same DEM sequences.
- DM03, because DM03 clears all the currently active DTCs while DM01 reports all the currently active DTCs).
- DM06, because DM01 reports all the OBD-relevant pending DTCs.
- DM12, because DM01 reports all the OBD-relevant active DTCs.
- DM23, because they use the same DEM sequences.

### 3.4.2 Queuing Mechanism

In general, no queuing mechanism is supported, so when a request is received and it is not possible to process it, it is rejected. The exception to this is for the DM02, DM03, DM06, DM12 or DM23 services, which do have a queuing feature.

A periodic/Dem triggered DM01 request will be queued but processed only after the previous DM02, DM03, DM06, DM12 or DM23 request is completed, or TxConfirmation is received for the same.

### 3.4.3 DTC Status Filter

The DTC status filter DEM_J1939DTC_PERMANENT is not supported.

### 3.4.4 SWS_J1939Dcm_00146

This requirement dictates that when a `J1939Dcm_TxConfirmation` call is received, the state of the channel has to be checked. This requirement is not taken into consideration.

### 3.4.5 Priority

The Priority which is set during the transmission of DM messages is the default priority as specified for each DM as per J1939SAE-73.The priority given by lower layers is ignored and not considered.

### 3.4.6 Miscellaneous

- J1939Dcm will not work with AUTOSAR 4.0 Dem because this version does not support the necessary `J1939Dcm_Dem` interfaces. Therefore, the same should be stubbed by the product lines.
- The Project must ensure that DM01 is configured only in one node and on a separate channel on the lower layer.
- The attribute *MetaDataType* is currently not supported. Therefore, only one *MetaDataItem* element should be configured per PDU. In case more than one *MetaDataItem* is configured, only the Metadata length of the first element will be fetched by J1939Dcm.

## 3.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- J1939Dcm/J1939DcmConfigSet/J1939DcmChannel
- J1939Dcm/J1939DcmConfigSet/J1939DcmDspExternalSRDataElementClass
- J1939Dcm/J1939DcmConfigSet/J1939DcmDspExternalSRDataElement-Class/J1939DcmDataElementInstance
- J1939Dcm/J1939DcmConfigSet/J1939DcmDspExternalSRDataElement-Class/J1939DcmDataElementInstance/J1939DcmDataElementInstanceRef
- J1939Dcm/J1939DcmConfigSet/J1939DcmDspExternalSRDataElement-Class/J1939DcmSubElementInDataElementInstance
- J1939Dcm/J1939DcmConfigSet/J1939DcmDspExternalSRDataElement-Class/J1939DcmSubElementInDataElementInstance/J1939DcmSubElementInDataElementInstanceRef
- J1939Dcm/J1939DcmConfigSet/J1939DcmDspExternalSRDataElement-Class/J1939DcmSubElementInImplDataElementInstance
- J1939Dcm/J1939DcmConfigSet/J1939DcmDspExternalSRDataElement-Class/J1939DcmSubElementInImplDataElementInstance/J1939DcmSubElementInImplDataElementInstanceRef
- J1939Dcm/J1939DcmConfigSet/J1939DcmProcessingConditions/J1939Dcm-ModeCondition/J1939DcmSwcSRDataElementValue/J1939DcmSwcSR-DataElementArray
- J1939Dcm/J1939DcmConfigSet/J1939DcmProcessingConditions/J1939Dcm-ModeCondition/J1939DcmSwcSRDataElementValue/J1939DcmSwcSR-DataElementArray/J1939DcmSwcSRDataElementArrayElement
- J1939Dcm/J1939DcmConfigSet/J1939DcmProcessingConditions/J1939Dcm-ModeCondition/J1939DcmSwcSRDataElementValue/J1939DcmSwcSR-DataElementArray/J1939DcmSwcSRDataElementArrayElement/J1939Dcm-SwcSRDataElementArrayElementIndex
- J1939Dcm/J1939DcmConfigSet/J1939DcmProcessingConditions/J1939Dcm-ModeCondition/J1939DcmSwcSRDataElementValue/J1939DcmSwcSR-DataElementArray/J1939DcmSwcSRDataElementArrayElement/J1939Dcm-SwcSRDataElementArrayElementValue
- J1939Dcm/J1939DcmConfigSet/J1939DcmProcessingConditions/J1939Dcm-ModeCondition/J1939DcmSwcSRDataElementValue/J1939DcmSwcSR-DataElementPrimitive
- J1939Dcm/J1939DcmConfigSet/J1939DcmProcessingConditions/J1939Dcm-ModeCondition/J1939DcmSwcSRDataElementValue/J1939DcmSwcSR-DataElementPrimitive/J1939DcmSwcSRDataElementPrimitiveValue
- J1939Dcm/J1939DcmConfigSet/J1939DcmProcessingConditions/J1939Dcm-ModeRule/J1939DcmModeRuleNrcValue

- J1939Dcm/J1939DcmGeneral/J1939DcmDM35BufferSize
- J1939Dcm/J1939DcmGeneral/J1939DcmMaxTransmitRetries

## 3.6 Deviations

### 3.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

#### 3.6.1.1 J1939Dcm

| | |
|---|---|
| [RTA-BSW] Description | Configuration of J1939Dcm (J1939 Diagnostic Communications Manager) module. |
| [AUTOSAR] Description | The SAE J1939 Dcm module |

#### 3.6.1.2 J1939Dcm/J1939DcmConfigSet/J1939DcmProcessingConditions/J1939DcmModeCondition/J1939DcmSwcSRDataElementRef

| | |
|---|---|
| [RTA-BSW] Upper Multiplicity | 0 |
| [AUTOSAR] Upper Multiplicity | 1 |

#### 3.6.1.3 J1939Dcm/J1939DcmConfigSet/J1939DcmProcessingConditions/J1939DcmModeCondition/J1939DcmSwcSRDataElementValue

| | |
|---|---|
| [RTA-BSW] Multiplicity | 0..0 |
| [AUTOSAR] Multiplicity | 1..1 |

#### 3.6.1.4 J1939Dcm/J1939DcmGeneral/J1939DcmDM01MaxDTCs

| | |
|---|---|
| [RTA-BSW] Description | Configuration value of limitation of maximum DTCs to be reported in the DM01 message.This parameter is applicable only for DEM triggered and Cyclically triggered DM01 message.For Tester triggered DM01 messages this limitation is not applicable. |
| [AUTOSAR] Description | Configuration value of limitation of maximum DTCs to be reported in the DM01 message. |

## 3.7 API Definition

Table 3.15.J1939Dcm Supported API functions

| | |
|---|---|
| `J1939Dcm_MainFunction` | Main function of the J1939 Diagnostic Communication Manager used for scheduling purposes and timeout supervision. |
| `J1939Dcm_Init` | Initializes the J1939 Diagnostic Communication Manager. |

| | |
|---|---|
| `J1939Dcm_DeInit` | Resets the J1939 Diagnostic Communication Manager to the uninitialized state. |
| `J1939Dcm_GetVersionInfo` | Returns the version information of the J1939 Diagnostic Communication Manager. |
| `J1939Dcm_SetState` | Changes the communication state of J1939Dcm to offline or online. |
| `J1939Dcm_RequestIndication` | Indicates reception of a Request PG. |
| `J1939Dcm_RxIndication` | Indication of a received I-PDU from a lower layer communication interface module. |
| `J1939Dcm_TxConfirmation` | The lower layer communication interface module confirms the transmission of an IPDU. |
| `J1939Dcm_StartOfReception` | Called at the start of receiving an N-SDU. |
| `J1939Dcm_CopyRxData` | Called to provide the received data of an I-PDU segment (N-PDU) to the upper layer. |
| `J1939Dcm_TpRxIndication` | Called after an I-PDU has been received via the TP API and indicates whether transmission was successful or not. |
| `J1939Dcm_CopyTxData` | Called to acquire the transmit data of an I-PDU segment (N-PDU). |
| `J1939Dcm_TpTxConfirmation` | Called after the I-PDU has been transmitted on its network and indicates whether transmission was successful or not. |
| `J1939Dcm_DemTriggerOnDTCStatus` | Trigger for DM01 message that a DTC status change has happened. |
| `J1939Dcm_GenericDMxTransmit` | Requests the transmission of a diagnostic message. |

For further information on these functions, please refer to AUTOSAR_SWS_SAE-J1939DiagnosticCommunicationManager.pdf

## 3.8 Configuration Advice

**NOTE**

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the “Configuration Generation” section of the RTA-CAR User Guide.

### 3.8.1 General

Each DM message can only be configured once in a particular node.

If DM04 or DM25 is configured, then the parameter

`J1939DcmMaxFreezeFramesPerMainFunction` should also be configured.

If DM14 is configured, DM15 and DM16 should also be configured.

If DM14, DM15 and DM16 (and DM18) are configured, then the following parameters should also be configured:

- `J1939DcmRbMemMaxNumberOfRetryForTransmit`
- `J1939DcmRbMemTimeOutForData`
- `J1939DcmRbMemTimeOutForOperationComplete`
- `J1939DcmRbMemTimeOutForOperationExecution`
- `J1939DcmRbMemTimeOutForReceivingKey`

### 3.8.2 J1939DcmRxPdu and J1939DcmTxPdu

For DM01, DM02, DM04, DM05, DM06, DM12, DM15, DM19, DM20, DM21, DM23, DM24, DM25, DM26 and DM31 the parameter `J1939DcmRxPdu` should not be configured and the parameter `J1939DcmTxPdu` should be configured.

For DM03 and DM05 neither `J1939DcmTxPdu` nor `J1939DcmRxPdu` should be configured.

For DM13 and DM14 only `J1939DcmRxPdu` should be configured.

For DM16 and DM18 both `J1939DcmRxPdu` and `J1939DcmTxPdu` should be configured.

Metadata length for all the J1939Dcm Pdu's (Rx/Tx path) should be 4 bytes.

### 3.8.3 DM01 Configuration

DM01 can be configured only in one node.

DM01 has to be transmitted periodically every 1s. This 1s mark can be altered when requesting DM01 is not feasible (e.g. StateMachine is busy) at the end of that 1s (via J1939Dcm_DM1Scheduler).

If DM01 is configured, then the parameter `J1939DcmDM01BufferSize` should be configured with a minimum value of 4 * `J1939DcmMaxDTCsPerMainFunction`.

If DM01 is not configured, then the parameter `J1939DcmDM01BufferSize` should not be configured.

#### 3.8.3.1 J1939DcmDM01MaxDTCs

`J1939DcmDM01MaxDTCs` should only be configured if `J1939DcmDmxSupport` is set to J1939DCM_DM01_SUPPORT for at least one `J1939DcmDiagnosticMessageSupport`.

### 3.8.4 DM18 Configuration

DM18 is optional if Long Seed/Key Security is to be used.

### 3.8.5 DM19 Configuration

If DM19 is configured, then the project should support RTE.

### 3.8.6 DM24 Configuration

The parameter `J1939DcmSPNsInDataStream` should only be configured if DM24 is configured.

### 3.8.7 Stopping DM01 Periodic Message

A new configuration parameter `J1939DcmRbAllowApplToStopDM1PeriodicMsg` is provided to enable/disable the DM01 Periodic Message feature. This parameter is optional and should be configured only if DM01 is supported in the project.

If `J1939DcmRbAllowApplToStopDM1PeriodicMsg` is set to TRUE, then the Application callback API `J1939DcmAppl_AllowDM1PeriodicMsg()` will be invoked to allow the application to either stop or allow the DM01 periodic messages for the current cycle. This application call will be invoked from `J1939Dcm_DM1Scheduler` and `J1939Dcm_DemTriggerOnDTCStatus` API after all the required checks are passed.

If there is no requirement to stop the DM01 periodic messages on certain conditions, then this feature can be disabled by setting `J1939DcmRbAllowApplToStopDM1PeriodicMsg` to FALSE.

### 3.8.8 DM01 Broadcasting on Multiple Nodes

The parameter `J1939DcmRbDM01MultiNodeResponse` is optional and should be set to TRUE only if DM01 is supported, else a validation error will be reported.

If `J1939DcmRbDM01MultiNodeResponse` is TRUE, then the callback API `J1939Dcm_DM1_GetBroadcastNode` will be invoked, allowing the Application to retrieve the NodeId on which DM01 will be transmitted. This NodeId can be different to the one on which DM01 is available in the current configuration.

If the Application has no requirement to change the NodeId from that configured for DM01 to be transmitted, then this feature can be disabled by setting `J1939DcmRbDM01MultiNodeResponse` to FALSE. In this case, the Application call will not be invoked.

### 3.8.9 Disabling the acceptance/rejection of new DM request Messages

If the application has no requirement to accept or reject new DM request messages on certain conditions, the `J1939DcmRbGetPermToAcceptReq` configuration extension parameter can be used to disable this functionality.

This parameter is optional and should be configured only if the application wants to accept or reject new DM request messages.

If `J1939DcmRbGetPermToAcceptReq` is set to TRUE, the Application callback API (J1939DcmAppl_GetPermToAcceptReq) will be invoked to allow the application to either accept or reject new DM request messages whenever a new request is received. This application call will be invoked from the J1939Dcm_StartOfReception, J1939Dcm_RequestIndication and J1939Dcm_RxIndication APIs after all the required checks are passed.

If `J1939DcmRbGetPermToAcceptReq` is set to FALSE (which is the default), the Application call will not be invoked.

## 3.9 Integration Advice

### 3.9.1 Init Functions

J1939Dcm_Init must be called by all projects that use J1939Dcm.

- J1939Dcm initializes the corresponding configuration pointer.
- All the configured channels are set to Offline State.
- The Common RAM structure and DM01 RAM structure are initialized to default values.
- The J1939Dcm Init global enum variable is updated to Initialized state.

### 3.9.2 DeInit Functions

J1939Dcm_Shutdown shuts down the module.

### 3.9.3 Scheduled Functions

The only J1939Dcm entity that has to be scheduled is J1939Dcm_MainFunction.

#### 3.9.3.1 Setting the Channel state to online state

Before any DM service is requested, the Channel used by that particular DM service should be set to Online state. This can be done by calling the J1939Dcm_SetState API to request that the state of the channel be updated to Online.

Pass the required channel, node and new state parameter J1939DCM_STATE_ONLINE to set online state -

- J1939Dcm_SetState( channel, node, J1939DCM_STATE_ONLINE )

#### 3.9.3.2 J1939Dcm_DemTriggerOnDTCStatus

To support the `J1939Dcm_DemTriggerOnDTCStatus` API, the Dem configuration parameter DemCallbackDTCStatusChanged must be configured. Implement your own callback function with your own condition (eg: "only 1 state change per DTC in one second is taken into account", etc) and call the code:*J1939Dcm_DemTriggerOnDTCStatus* API.

#### 3.9.3.3 Generic DMx Support

The flow of a DM request/response to/from an Application is explained below and can help in the development of such interfaces:

On reception of a Request or Request2 PG for an unsupported diagnostic message, the `GenericDMxRequestIndication` function will be called by J1939Dcm. After processing the DM, the Application will inform J1939Dcm by calling `J1939Dcm_GenericDMxTransmit`. J1939Dcm calls `GenericDMxCopyTxData`, with `SduLength` = 0, to query the current amount of available data in the Application. J1939Dcm then calls `GenericDMxCopyTxData` again, with `SduLength` now set to the value of available data in the Application. The Application updates `SduDataPtr` with

the response and sets `availableDataPtr` to 0. J1939Dcm then transmits the response and confirms it by calling `GenericDMxTxConfirmation`.

When a request is received via J1939Tp for an unsupported DM, J1939Dcm calls `GenericDMxStartOfReception`. This call indicates to the Application that an unsupported DM request is incoming and also provides its total length. J1939Dcm transfers the request data in one single call by calling `GenericDMxCopyRxData`. Finally, it calls `GenericDMxRxIndication` to indicate whether or not reception was successful.

### 3.9.4 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The J1939Dcm module uses the following OS resources:

- J1939Dcm_Global_NoNest

## 3.10 Hardware Requirements

Table 3.16.J1939Dcm Hardware Requirements

| Support | Usage |
|---|---|
| FLASH | 3970 bytes |
| RAM | 2931 bytes |

Usage calculated based on a sample configuration.

# 4 J1939 Request Manager

| | |
|---|---|
| Module name | J1939Rm |
| Package | RTA-J1939 |
| AUTOSAR Module ID | 59 |
| AUTOSAR Version | 22-11 |

## 4.1 Introduction

The J1939 Request Manager (J1939Rm) is part of the Communication Stack and it deals with the *Request-Response-based PGN's* of SAE J1939.

On one side, J1939Rm is responsible for routing incoming requests to the correct destination, and providing an infrastructure for sending responding *ACKM PGs*.

On the other side, J1939Rm also provides an infrastructure to send *RQST PGs*, and to supervise timeout of the response(s), including (but not limited to) ACKM.

The following diagrams summarise the Reception of *Request PGs* and the Transmission of Non-Diagnostic messages:

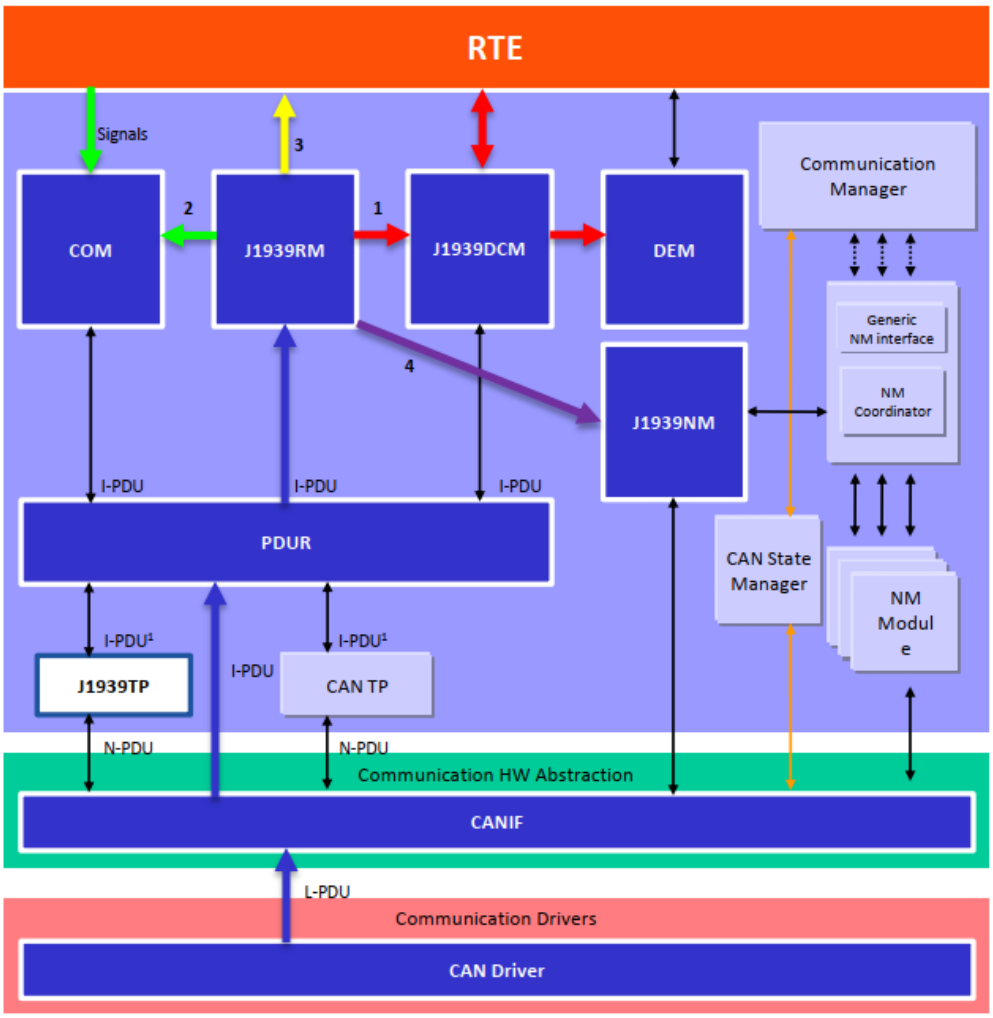


Figure 4.1.Reception of request PG

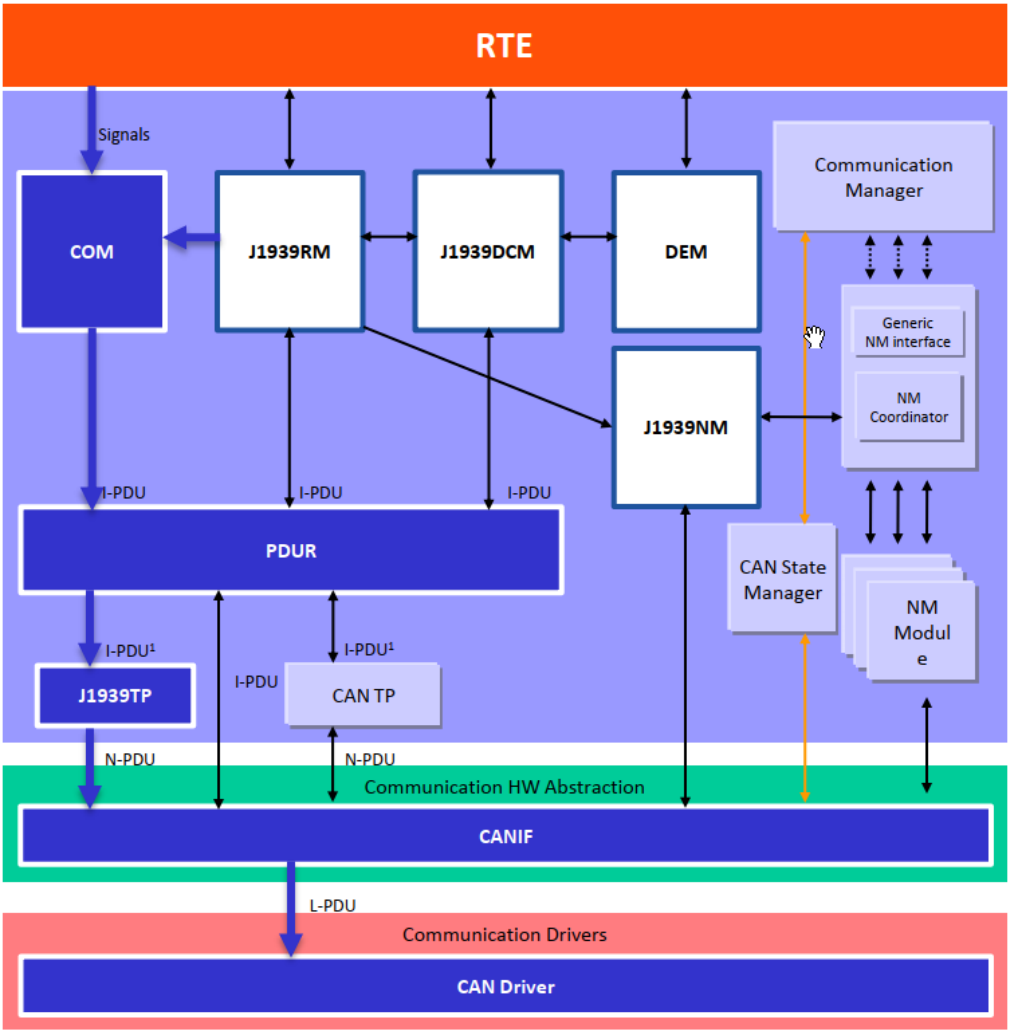


Figure 4.2.Transmission of Non-Diagnostic message

### 4.1.1 Interaction/Dependencies

The diagrams above show the position of J1939Rm (and the other J1939 modules) in the AUTOSAR layered architecture. The interaction/dependencies of J1939Rm with other modules is summarised here:

**COM**

- The COM module indicates the channel.

**J1939Nm**

- Provides the address.

**PduR**

- Provides the routing of J1939Rm to CanIf.

J1939Rm also interacts with J1939Dcm, J1939Tp, CAN Driver, CanIf, Det, Dem and application software components (SW-Cs) via the RTE. It also supports Complex Drivers (CDD).

**Module Configuration**

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

## 4.2 Supported Features

J1939Rm supports the following features.

### 4.2.1 Reception of Request

J1939Rm receives the requests for a PGN and forwards it to a specific user or to all configured users, depending on whether the request is specific or broadcast. In this context, a user can be J1939Dcm, J1939Nm, Com or CDD. (RTE as a user is not supported). If the user is Com, J1939Rm triggers the response to the request.

### 4.2.2 Reception of Acknowledgement

J1939Rm receives the acknowledgement and forwards it to the corresponding users. In this context, the user can be CDD and RTE.

### 4.2.3 Transmission of Request

J1939Rm provides an API (J1939Rm_SendRequest) to transmit a request from a user. A channel level queue handles parallel request transmissions. In this context, the user is CDD or RTE.

### 4.2.4 Transmission of Acknowledgement

J1939Rm provides an API (J1939Rm_SendAck) to transmit an acknowledgement from a user. A channel level queue handles parallel acknowledgment transmissions. In this context, users can be J1939Dcm, CDD or RTE.

### 4.2.5 Pgn Calibration Support

The J1939RmSupportRequestPgnCalibration and J1939RmSupportAcknowledgementPgnCalibration extension parameters can be used to enable/disable calibration for request and acknowledgement for all users, including users that support multiple Pgn (i.e. Request and Acknowledgement).

When two users support the same Pgn, it is represented by diferent a variable. When calibration is enabled, the request is sent to one user at a time. Note that J1939Dcm does not support the calibration feature, so you should not use calibration for Dcm users.

### 4.2.6 Post-Build Selectable Variant Support

This feature applies for some of the parameters.

### 4.2.7 Reception of Multi-PG Request

- Multi-PG protocol provides a transport mechanism to carry multiple SAE J1939 PGs in a single CAN FD data frame. By default Multi-PG Feature is disabled in J1939Rm configuration ie. J1939RmMultiPgSupport set as false.
- When sending a Destination Specific FEFF Multi-PG data frame, all the included PGs must be Destination Specific PGs directed to the same Destination Address specified in the CAN identifier field.
- The users can be CDD, RTE or J1939DCM.
- The Multi-PG Service will be supported for only RQST PG and Max 9 C-PGN's.

### 4.2.8 Routing of Request for J1939Nm AC PG

- To activate the Gateway feature the J1939RmGatewaySupport must be enabled in the J1939Rm configuration.
- When J1939Rm GatewaySupport is enabled and J1939RmChannel is linked to another J1939RmChannel via a J1939NmSharedAddressSpace, All J1939Nm AddressClaim PG received on first channel shall be forwarded to the Second Channel/Connected Shared Channel.
- The user can be J1939Nm.
- The J1939 Request Manager Shall Route Incoming Request PG and Acknowledgement PG to Connected Channels.

## 4.3 Extensions

### 4.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

#### 4.3.1.1 J1939Rm/J1939RmGeneral/J1939RmMultiPgSupport

| | |
|---|---|
| Short Name | J1939RmMultiPgSupport |
| Description | Enables/Disables Support for Multi-Pg Feature. |
| Multiplicity | 1..1 |
| Default Value | - |
| Type | ECUC-BOOLEAN-PARAM-DEF |

#### 4.3.1.2 J1939Rm/J1939RmGeneral/J1939RmSupportAcknowledgementPgnCalibration

| | |
|---|---|
| Short Name | J1939RmSupportAcknowledgementPgnCalibration |
| Description | Pre-processor switch for enabling support of Acknowledgement PGN calibration. |

| | |
|---|---|
| Multiplicity | 0..1 |
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

#### 4.3.1.3 J1939Rm/J1939RmGeneral/J1939RmSupportChannelCalibration

| | |
|---|---|
| Short Name | J1939RmSupportChannelCalibration |
| Description | Pre-processor switch for enabling support of request PGN Channel calibration. |
| Multiplicity | 0..1 |
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

#### 4.3.1.4 J1939Rm/J1939RmGeneral/J1939RmSupportRequestPgnCalibration

| | |
|---|---|
| Short Name | J1939RmSupportRequestPgnCalibration |
| Description | Pre-processor switch for enabling support of request PGN calibration.<br>J1939DCM user will not work with the calibration |
| Multiplicity | 0..1 |
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

#### 4.3.1.5 J1939Rm/J1939RmGeneral/J1939RmTxConfirmationTimeout

| | |
|---|---|
| Short Name | J1939RmTxConfirmationTimeout |
| Description | Time in seconds to wait for a confirmation after transmission of a message.<br>The behaviour when the time elapses depends on the transmitted message. |
| Multiplicity | 1..1 |
| Default Value | - |
| Type | ECUC-FLOAT-PARAM-DEF |

#### 4.3.1.6 J1939Rm/J1939RmGeneral/J1939RmUserCallout

| | |
|---|---|
| Short Name | J1939RmUserCallout |
| Description | Pre-processor switch for enabling support of User Callouts. |
| Multiplicity | 0..1 |
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

### 4.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

- There are no additional API defined in this version.

## 4.4 Limitations

- Compiler abstraction support is deprecated, henceforth the component has to be used only for compilers >= 32-bit controllers.
- J1939Dcm doesn't support the calibration feature.

## 4.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- J1939Rm/J1939RmConfigSet/J1939RmChannel/J1939RmRequestQueue2-Size
- J1939Rm/J1939RmConfigSet/J1939RmChannel/J1939RmRqst2RxPdu
- J1939Rm/J1939RmConfigSet/J1939RmChannel/J1939RmRqst2RxPdu/J1939RmRqst2RxPduId
- J1939Rm/J1939RmConfigSet/J1939RmChannel/J1939RmRqst2RxPdu/J1939RmRqst2RxPduRef
- J1939Rm/J1939RmConfigSet/J1939RmChannel/J1939RmRqst2TxPdu
- J1939Rm/J1939RmConfigSet/J1939RmChannel/J1939RmRqst2TxPdu/J1939RmRqst2TxPduId
- J1939Rm/J1939RmConfigSet/J1939RmChannel/J1939RmRqst2TxPdu/J1939RmRqst2TxPduRef
- J1939Rm/J1939RmConfigSet/J1939RmNode/J1939RmUser/J1939RmComUser/J1939RmComIPdu/J1939RmComIPduExtId1
- J1939Rm/J1939RmConfigSet/J1939RmNode/J1939RmUser/J1939RmComUser/J1939RmComIPdu/J1939RmComIPduExtId2
- J1939Rm/J1939RmConfigSet/J1939RmNode/J1939RmUser/J1939RmComUser/J1939RmComIPdu/J1939RmComIPduExtId3
- J1939Rm/J1939RmGeneral/J1939RmSupportRequest2

## 4.6 Deviations

### 4.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

#### 4.6.1.1 J1939Rm/J1939RmConfigSet/J1939RmNode/J1939RmUser/J1939RmCddUser/J1939RmUserId

| | |
|---|---|
| [RTA-BSW] Maximum Value | 65535 |
| [AUTOSAR] Maximum Value | 254 |

#### 4.6.1.2 J1939Rm/J1939RmConfigSet/J1939RmNode/J1939RmUser/J1939RmDcmUser/J1939RmUserId

| | |
|---|---|
| [RTA-BSW] Maximum Value | 65535 |
| [AUTOSAR] Maximum Value | 254 |

#### 4.6.1.3 J1939Rm/J1939RmConfigSet/J1939RmNode/J1939RmUser/J1939RmRteUser/J1939RmUserId

| | |
|---|---|
| [RTA-BSW] Maximum Value | 65535 |
| [AUTOSAR] Maximum Value | 254 |

#### 4.6.1.4 J1939Rm/J1939RmGeneral/J1939RmGatewaySupport

| | |
|---|---|
| [RTA-BSW] Description | Enables/Disables Support for Routing Request and Acknowledgement messages. |
| [AUTOSAR] Description | Enables/disables support for routing Request and Acknowledgement messages. |

## 4.7 API Definition

Table 4.12.J1939Rm Supported API functions

| | |
|---|---|
| `J1939Rm_MainFunction` | Main function of the J1939 Request Manager used for scheduling purposes and timeout supervision. |
| `J1939Rm_Init` | Initializes the J1939 Request Manager. |
| `J1939Rm_DeInit` | Resets the J1939 Request Manager to the uninitialized state. |
| `J1939Rm_GetVersionInfo` | Returns the version information of the J1939 Request Manager. |
| `J1939Rm_SetState` | Changes the communication state of J1939Rm to offline or online. |
| `J1939Rm_SendRequest` | Requests transmission of a Request PG. |
| `J1939Rm_CancelRequestTimeout` | Cancels timeout monitoring of a Request. |
| `J1939Rm_SendAck` | Requests transmission of an Acknowledgement PG. |
| `J1939Rm_RxIndication` | Indication of a received I-PDU from a lower layer communication interface module. |

| | |
|---|---|
| `J1939Rm_TxConfirmation` | The lower layer communication interface module confirms the transmission of an IPDU. |
| `J1939Rm_CheckReceivedComIPdu` | Reports a received ComIPdu. |

For further information on these functions, please refer to AUTOSAR_SWS_SAE-J1939RequestManager.pdf

## 4.8 Configuration Advice

> **NOTE**
>
> Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

### 4.8.1 Initialization

Configure `J1939Rm_Init` in EcuMDriverInitListOne in order to initialize J1939Rm.

### 4.8.2 Metadata Support

Metadata support must be enabled in CanIf (CanIfMetaDataSupport = True).

If any of the users is Com, Metadata support should also be enabled in Com (ComMetaDataSupport = True)

For all the J1939Rm PDUs, Metadata should be configured as 4bytes in EcuC.

### 4.8.3 Using Metadata Directly as CanId of J1939 messages

To enable Metadata to be used directly as CanId, the following configuration is required.

- MetaDataLength should be >= Extended CanID (as noted above, typically it is set to 4 in the relevant Pdu configuration in the EcuC module).
- ComMetaDataDefault should be configured, two examples of which are shown here.

Example 1: 1,255,255,7

- 1 = [8-bit Source Address]
- 255 = [8-bit PDU-Specific]
- 255 = [8-bit PDU-Format]
- 7 = [5-bit Priority and Data Page]

Example 2: 1,2,202,7

- 1 = [8-bit Source Address]

- 2 = [8-bit PDU-Specific]
- 202 = [8-bit PDU-Format]
- 7 = [5-bit Priority and Data Page]

If, as in example 1, PDU Format is >= 240 (PDU2 Format), the PDU-Specific is part of the requested PG.

If, as in example 2, PDU Format is <240 (PDU1 Format), then PDU-Specific is the Destination Address.

`CanIfTxPduCanId` and `CanIfTxPduCanIdMask` of the relevant CanIf Txpdu should be left blank, and in this case J1939Rm will just use information from MetadataDefault, or forwarded Metadata directly, as CanIf.

Note: If you do configure `CanIfTxPduCanId` and `CanIfTxPduCanIdMask` of the relevant CanIf Txpdu, J1939Rm will use this information and Metadata info to calculate the CanId.

- `ComTxIPduDeferredProcessing` should be set to false.

### 4.8.4 Node and Channel Referencing

All nodes referenced by J1939Rm must be present in J1939Nm, and the reference to each J1939Nm node by J1939Rm must be unique - a node defined in J1939Nm can only be referenced by one node in J1939Rm.

All channels referenced by J1939Rm must be present in J1939Nm.

## 4.9 Integration Advice

### 4.9.1 Init Functions

`J1939Rm_Init` is called by EcuM during the system initialization

### 4.9.2 DeInit Functions

`J1939Rm_DeInit` de-initializes the module.

### 4.9.3 Scheduled Functions

`J1939Rm_MainFunction` is called for scheduling purposes and timeout supervision. The frequency of invocations of J1939Rm_MainFunction is determined by the configuration parameter J1939RmMainFunctionPeriod.

### 4.9.4 Misc

J1939Rm must be integrated along with the ComStd package (the Autosar Communication Stack). At least J1939Nm (if you're planning to use it) and CanIf with metadata support must be integrated with J1939Rm.

The node and channels configured in J1939Nm will be referenced from J1939Rm.

BswM needs to be configured to change the node-channel states.

If you're integrating J1939Nm with J1939Rm, set all node-channel states of

J1939Rm to J1939RM_STATE_ONLINE.

### 4.9.5 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The J1939Rm module uses the following OS resources:

- J1939Rm_SetStateNoNest
- J1939Rm_WriteToTimeoutBuffer

## 4.10 Hardware Requirements

Table 4.13.J1939Rm Hardware Requirements

| Support | Usage |
|---|---|
| FLASH | 1743 bytes |
| RAM | 28 bytes |

Usage calculated based on a sample configuration.

# 5 J1939 Network Management

| | |
|---|---|
| Module name | J1939Nm |
| Package | RTA-J1939 |
| AUTOSAR Module ID | 34 |
| AUTOSAR Version | 22-11 |

## 5.1 Introduction

The J1939 Network Management module (J1939Nm) is part of the Communication Stack and is responsible for assigning a unique address to each ECU.

It supports the transmission and reception of *AddressClaimed PGs*, and the handling of requests for the AddressClaimed PGs (ACs).

It also supports transmission of ACs on start-up, after a contending AC is received from another node, and on request (triggered by J1939Rm, the J1939 Request Manager).

J1939Nm also ensures that the ECU does not send messages during the initial address claiming phase or after the ECU has sent a *CannotClaimAddress PG* because it has lost its address to a contending Address Claim.

The following diagrams summarise the Reception of AddressClaimed PGs and the Transmission of AddressClaimed PGs:

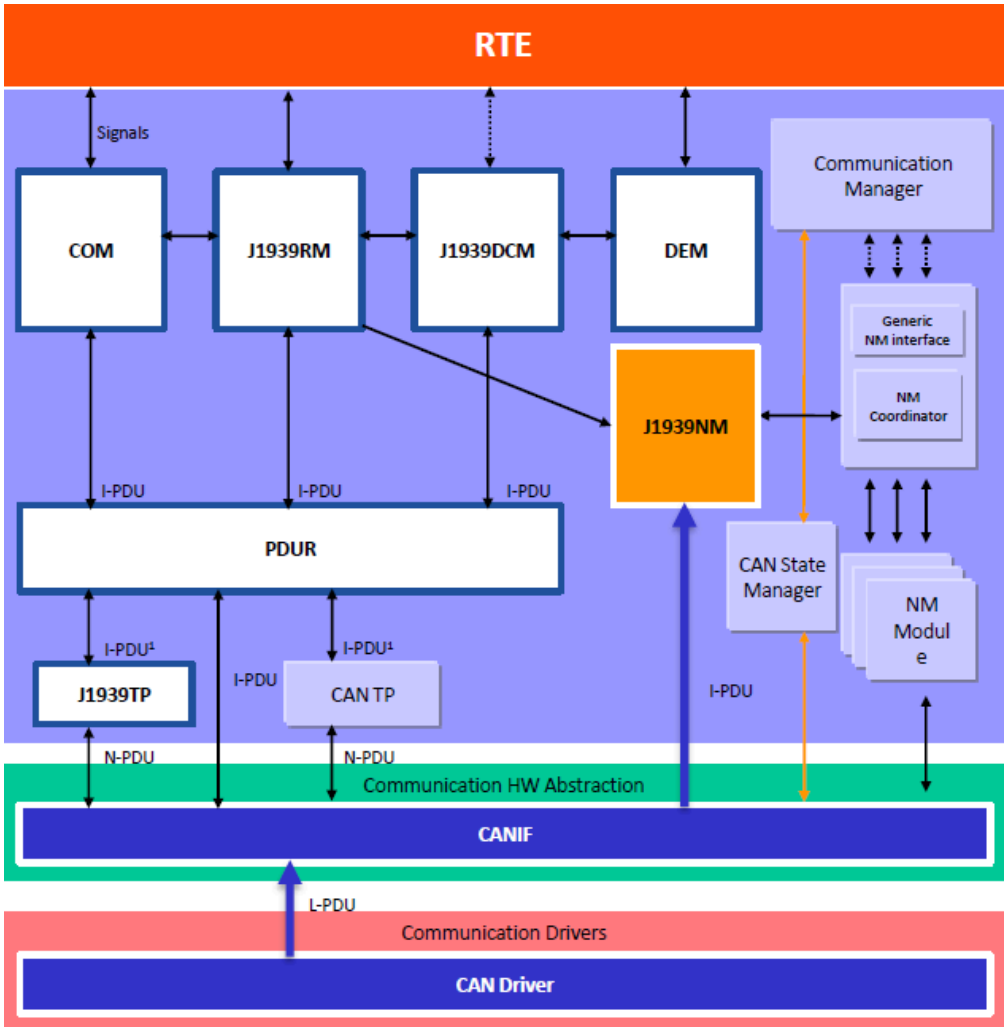


Figure 5.1.Reception of AddressClaimed PG

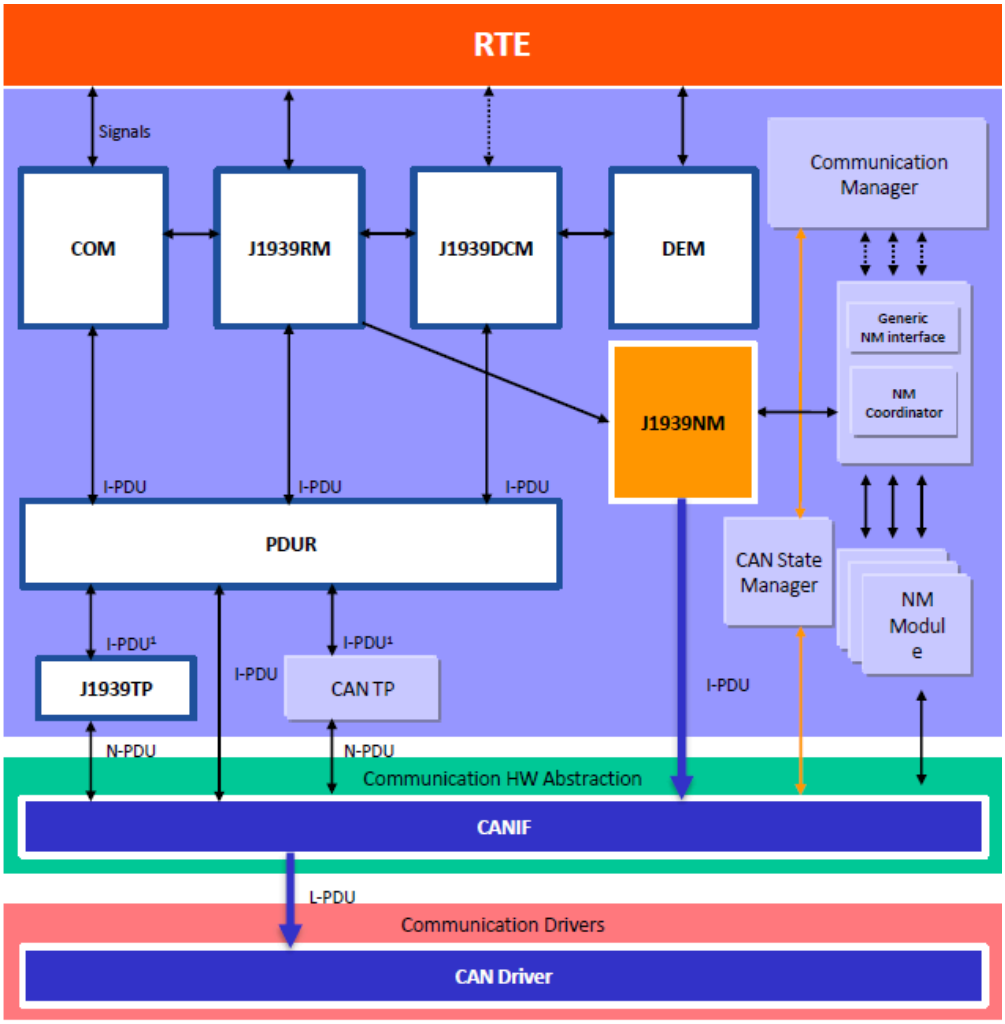


Figure 5.2.Transmission of AddressClaimed PG

### 5.1.1 Interaction/Dependencies

The diagrams above show the position of J1939Nm (and the other J1939 modules) in the AUTOSAR layered architecture. The interaction/dependencies of J1939Nm with other modules is summarised here:

**J1939Rm**

- J1939Rm requests the J1939Nm module to send the AddressClaim PG.

**BswM**

- J1939Nm calls `BswM_J1939Nm_StateChangeNotification` to inform of state changes for every Node Channel.

**Nm**

- J1939Nm provides *Nm Mode state* and *Nm Channel state* whenever there is a corresponding change of state. These notifications are consolidated for

each J1939 Channel (i.e. When one *NodeChannel* moves to *Network Mode*, Nm is notified. When all the *NodeChannels* are in *AcLost* then Nm is notified).

**CanIf**

- AddressClaim PGs and CannotClaimAddress PGs are triggered onto the bus by the J1939Nm module via `CanIf_Transmit`.
- AddressClaim PGs are indicated to the J1939Nm module via `J1939Nm_RxIndication`. Confirmations of transmitted AddressClaim PGs are provided by `J1939Nm_TxConfirmation`.

**Other**

- J1939Nm also has interactions with the J1939Dcm, Det and Dem modules.

**Module Configuration**

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

## 5.2 Supported Features

- Transmission and reception of AddressClaim PGs (and CannotClaimAddress) after ECU initialization.
- Request for Address Claimed. This is used to determine available addresses and to find out which ECUs are currently on the network.
- Provides an interface to J1939Rm to request the transmission of AddressClaim PG.
- Provides consolidated state change notifications of a particular J1939Nm channel to NmIf and BswM.
- Provides the value of Bus-off delay to CanSm when requested via prescribed Callout.
- Dynamic/Arbitrary Address Claim support.
- Reclaim address support.
- XCP calibration support.

## 5.3 Extensions

### 5.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

#### 5.3.1.1 J1939Nm/J1939NmConfigSet/J1939NmChannel/J1939NmNodeSpecificDemEventParameterRefs/J1939NmRbDynCreate_J1939NM_E_ADDRESS_LOST

| | |
|---|---|
| Short Name | J1939NmRbDynCreate_J1939NM_E_ADDRESS_LOST |

| | |
|---|---|
| Long Name | Dynamically create J1939NM_E_ADDRESS_LOST event |
| Description | Selects the way how the J1939Nm DemEvents are created in Dem and referenced in J1939Nm module. If not set or set to "Disabled", no such DemEvent is automatically/dynamically created in Dem. If the dynamic creation of this DemEvent is standard, then this DemEvent (in parameter J1939NM_E_ADDRESS_LOST) is dynamically created in Dem and referenced in J1939Nm module and hence shall not be set manually. |
| Multiplicity | 0..1 |
| Default Value | Disabled |
| Enumeration Literals | Disabled, Standard |
| Type | ECUC-ENUMERATION-PARAM-DEF |

#### 5.3.1.2 J1939Nm/J1939NmConfigSet/J1939NmChannel/J1939NmNumOfNodes

| | |
|---|---|
| Short Name | J1939NmNumOfNodes |
| Description | Number of nodes connected to the network to this J1939Nm channel. This value would help in deciding the number of address claim messages in "AC re-claim" state during arbitrary address claim. |
| Multiplicity | 0..253 |
| Value Range | 0..253 |
| Default Value | 1 |
| Type | ECUC-INTEGER-PARAM-DEF |

#### 5.3.1.3 J1939Nm/J1939NmGeneral/J1939NmBswmSupport

| | |
|---|---|
| Short Name | J1939NmBswmSupport |
| Description | Switches the notifications to BSWM ON or OFF. |
| Introduction | * true: enabled (ON). * false: disabled (OFF). |
| Multiplicity | 1..1 |
| Default Value | true |
| Type | ECUC-BOOLEAN-PARAM-DEF |

#### 5.3.1.4 J1939Nm/J1939NmGeneral/J1939NmCalibrationSupportEnabled

| | |
|---|---|
| Short Name | J1939NmCalibrationSupportEnabled |
| Description | Pre-processor switch for enabling the Calibration Support for J1939Nm. True : Calibration support is enabled False: Calibration support is disabled |
| Multiplicity | 0..1 |

| | |
|---|---|
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

### 5.3.1.5 J1939Nm/J1939NmGeneral/J1939NmCanIfStateCheck

| | |
|---|---|
| Short Name | J1939NmCanIfStateCheck |
| Description | Pre-processor switch for enabling the CanIf startup behavior check. True : CanIfstartup check Support is enabled therefore CanIf PduMode and Controller-Mode are checked before calling CanIf_Transmit False: CanIfstartup check Support is disabled and CanIf PduMode and ControllerMode are not checked before calling CanIf_Transmit |
| Multiplicity | 0..1 |
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

### 5.3.1.6 J1939Nm/J1939NmGeneral/J1939NmDSMReInitEnabled

| | |
|---|---|
| Short Name | J1939NmDSMReInitEnabled |
| Description | Enables or disables the reporting of DEM events to DSM after the Re-initialisation of DSM has occured. |
| Multiplicity | 0..1 |
| Default Value | false |
| Type | ECUC-BOOLEAN-PARAM-DEF |

### 5.3.1.7 J1939Nm/J1939NmGeneral/J1939NmMaximumNumberOfRetryEntries

| | |
|---|---|
| Short Name | J1939NmMaximumNumberOfRetryEntries |
| Description | This parameter is the maximum number of times J1939Nm retries sending out a Tx Pdu to canIf |
| Multiplicity | 0..1 |
| Value Range | 0..10 |
| Default Value | - |
| Type | ECUC-INTEGER-PARAM-DEF |

### 5.3.1.8 J1939Nm/J1939NmGeneral/J1939NmTxConfirmationTimeout

| | |
|---|---|
| Short Name | J1939NmTxConfirmationTimeout |
| Description | Time in seconds to wait for a confirmation after transmission of a message. The behaviour when the time elapses depends on the transmitted message. |

| | |
|---|---|
| Multiplicity | 1..1 |
| Value Range | 0..65.535 |
| Default Value | 1.5 |
| Type | ECUC-FLOAT-PARAM-DEF |

#### 5.3.1.9 J1939Nm/J1939NmGeneral/J1939NmUserCalloutHeaderFile

| | |
|---|---|
| Short Name | J1939NmUserCalloutHeaderFile |
| Description | Header file which is included by J1939Nm when J1939NmUserCallout is enabled. This header file must provide the prototype of the <User_AddressClaimedIndication> defined in J1939NmUserCallout. |
| Multiplicity | 0..1 |
| Default Value | - |
| Type | ECUC-STRING-PARAM-DEF |

### 5.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

#### 5.3.2.1 J1939Nm_GetNodeAddress

| | |
|---|---|
| Syntax | `J1939Nm_GetNodeAddress(uint8 NodeId)` |
| Param In | NodeId - Node by which the request was received. |
| Returns | I_dynamicAddress → Returns the address. |
| Description | Returns the updated address that has been changed through calibration. |

## 5.4 Limitations

- RTE as a user is not supported.
- Timeout supervision for transmitted requests is not supported.
- Compiler abstraction support is deprecated; use only for compilers >= 32 bit controllers.
- J1939NmExternalNode is not supported.
- When XCP calibration is enabled, runtime NodeId (address) changes are *not* supported because the Address Claimed PGs are packed at module initialization. In order to calibrate a different ECU node, therefore, the module must be reinitialized with its NodeId.

## 5.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- J1939Nm/J1939NmConfigSet/J1939NmExternalNode
- J1939Nm/J1939NmConfigSet/J1939NmExternalNode/J1939NmExternalNodeChannelRef
- J1939Nm/J1939NmConfigSet/J1939NmExternalNode/J1939NmExternalNodeGatewayedChannelRef
- J1939Nm/J1939NmConfigSet/J1939NmExternalNode/J1939NmExternalNodeId
- J1939Nm/J1939NmConfigSet/J1939NmExternalNode/J1939NmExternalNodeNameArbitraryAddressCapable
- J1939Nm/J1939NmConfigSet/J1939NmExternalNode/J1939NmExternalNodeNameECUInstance
- J1939Nm/J1939NmConfigSet/J1939NmExternalNode/J1939NmExternalNodeNameFunction
- J1939Nm/J1939NmConfigSet/J1939NmExternalNode/J1939NmExternalNodeNameFunctionInstance
- J1939Nm/J1939NmConfigSet/J1939NmExternalNode/J1939NmExternalNodeNameIdentityNumber
- J1939Nm/J1939NmConfigSet/J1939NmExternalNode/J1939NmExternalNodeNameIndustryGroup
- J1939Nm/J1939NmConfigSet/J1939NmExternalNode/J1939NmExternalNodeNameManufacturerCode
- J1939Nm/J1939NmConfigSet/J1939NmExternalNode/J1939NmExternalNodeNameVehicleSystem
- J1939Nm/J1939NmConfigSet/J1939NmExternalNode/J1939NmExternalNodeNameVehicleSystemInstance
- J1939Nm/J1939NmConfigSet/J1939NmExternalNode/J1939NmExternalNodePreferredAddress
- J1939Nm/J1939NmGeneral/J1939NmVersionInfoApi

## 5.6 Deviations

### 5.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

- There are no known deviations in this version.

## 5.7 API Definition

Table 5.12.J1939Nm Supported API functions

| | |
|---|---|
| `J1939Nm_MainFunction` | Main function of the J1939 Network Management module used for scheduling purposes and timeout supervision. |
| `J1939Nm_Init` | Initializes the J1939 Network Management Module. |
| `J1939Nm_DeInit` | Resets the J1939 Network Management Module to the uninitialized state. |
| `J1939Nm_GetVersionInfo` | Returns the version information of the J1939 Network Management Module. |
| `J1939Nm_NetworkRequest` | Request the network when the ECU needs to communicate on the bus. |
| `J1939Nm_NetworkRelease` | Release the network when the ECU doesn't have to communicate on the bus. |
| `J1939Nm_GetState` | Returns the state and the mode of the Network Management. |
| `J1939Nm_GetBusOffDelay` | Returns the number of CanSM base cycles to wait additionally to L1/L2 after a BusOff occurred. |
| `J1939Nm_PassiveStartUp` | Passive startup of the NM to trigger transition from Bus-Sleep Mode to the Network Mode without requesting the network. |
| `J1939Nm_RxIndication` | Indication of a received I-PDU from a lower layer communication interface module. |
| `J1939Nm_TxConfirmation` | The lower layer communication interface module confirms the transmission of an IPDU. |
| `J1939Nm_RequestIndication` | Indicates reception of a Request PG. |

For further information on these functions, please refer to AUTOSAR_SWS_SAE-J1939NetworkManagement.pdf

## 5.8 Configuration Advice

**NOTE**

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

### 5.8.1 General

To allow AddressClaim after ECU initialization, you need to set a value for the

parameter `J1939NmMaximumNumberOfRetryEntries`. The value should be greater than number of nodes configured in J1939Nm to avoid incorrect claimed address.

For Dynamic Address Claim support, a new sub-state J1939NM_STATE_AC_RE-CLAIM has been introduced to listen, store and select different address during address contention.

Configuration to the ComM channel is required from the J1939Nm Channel container.

Configuration to the EcuC Pdu is required, with a corresponding link to be provided in CanIf (for message transmission and reception).

If `J1939NmRbDynCreate_J1939NM_E_ADDRESS_LOST` is enabled, dynamic forwarding of DEM events will occur.

### 5.8.2 Source Address

The Source address is set by parameter `J1939NmNodePreferredAddress`, and it should not be the same as any other J1939 Nodes that are using same J1939Nm Channel.

### 5.8.3 Node and Channel Referencing

All nodes referenced by J1939Rm must be present in J1939Nm, and the reference to each J1939Nm node by J1939Rm must be unique - a node defined in J1939Nm can only be referenced by one node in J1939Rm.

All channels referenced by J1939Rm must be present in J1939Nm.

### 5.8.4 Network Management States of the J1939Nm

Assuming that `BswMJ1939NmEnabled` is set, J1939Nm reports state changes to the Basic Software Mode Manager (BswM). You can then configure BswM (Rule, Mode condition, etc…) to change the State of J1939Dcm and J1939Rm channel to ONLINE or OFFLINE.

### 5.8.5 XCP Calibration

Calibration with XCP is supported in the J1939 Stack. A call to the API `J1939Nm_GetNodeAddress` will return the node address that has been calibrated, upon passing the NodeId (the identifier of the node for which the request was received).

Both the J1939Rm and J1939Dcm modules can invoke the API to get the node address. If the NodeId is invalid (i.e. it does not match the NodeId of any ECU node), `J1939Nm_GetNodeAddress` returns *0xfeu* (i.e. Null Address).

To enable XCP calibration, set the parameter `J1939NmCalibrationSupportEnabled` to TRUE.

See Section 5.4 for more details on XCP calibration.

## 5.9 Integration Advice

### 5.9.1 Init Functions

`J1939Nm_Init` is called by EcuM during the system initialization

### 5.9.2 DeInit Functions

`J1939Nm_DeInit` de-initializes the module.

### 5.9.3 Scheduled Functions

`J1939Nm_MainFunction` is called for scheduling purposes and timeout supervision at a frequency determined by the configuration parameter J1939NmMainFunctionPeriod. Ensure that the value of J1939NmMainFunctionPeriod is in line with the Scheduling raster of the main function. This is independent of application and other functions.

### 5.9.4 Misc

J1939Nm must be integrated along with the ComStd package (the Autosar Communication Stack). At least J1939Rm, and CanIf with metadata support, must be integrated with J1939Nm.

### 5.9.5 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The J1939Nm module uses the following OS resources:

- J1939Nm_CommonLock

## 5.10 Hardware Requirements

Table 5.13.J1939Nm Hardware Requirements

| Support | Usage |
|---|---|
| FLASH | 2546 bytes |
| RAM | 92 bytes |

Usage calculated based on a sample configuration.

# 6 Glossary

**AC** J1939 AddressClaimed PG (PGN = 0x0EE00)

**ACK**
J1939 Acknowledgement PG (ACKM) with control byte set to 0

**ACKM**
J1939 Acknowledgement Message

**BAM**
Broadcast Announce Message, broadcast variant of SAE J1939 transport protocol

**BSW**
Basic Software (module)

**CA** Controller Application, role of an ECU tied to one address

**CMDT**
Connection Mode Data Transfer, peer-to-peer variant of SAE J1939 transport protocol

**DA** Destination Address, part of the 29 bit identifier of SAE J1939 messages

**DCM**
AUTOSAR Diagnostic Communication Manager

**DEM**
AUTOSAR Diagnostic Event Manager

**DET**
AUTOSAR Default Error Tracer

**DM** Diagnostic messages

**DP** Data Page, the most significant bit (MSB) of the 18 bit PGN

**EDP**
Extended Data Page, the second bit (after MSB) of the 18 bit PGN

**J1939Dcm**
AUTOSAR J1939 Diagnostic Communication Manager module

**J1939Nm**
AUTOSAR J1939 Network Management module

**J1939Rm**
AUTOSAR J1939 Request Manager module

**J1939Tp**
AUTOSAR J1939 Transport Layer module

**NACK**
J1939 Acknowledgement PG (ACKM) with control byte set to 1

**NAME**
Unique 64 bit identifier of each ECU connected to an SAE J1939 network

**Nm** AUTOSAR Network Management module

**Node**
J1939 node - can be attached to more than one channel

**NodeChannel**
The connection of a node to one channel

**PDUF**
PDU Format, part of the 29 bit identifier of SAE J1939 messages which identifies the message and determines the layout of the 29 bit identifier

**PduR**
AUTOSAR PDU Router

**PDUS**
PDU Specific, part of the 29 bit identifier of SAE J1939 messages which identifies broadcast messages which do not have a destination address

**PG** Parameter Group

**PGN**
Parameter Group Number, unique identifier of an SAE J1939 parameter group (contains EDP, DP, PDUF, PDUS)

**Rm** Request Manager

**RQST**
J1939 Request PG (PGN = 0x0EA00)

**RQST2**
J1939 Request2 PG (PGN = 0x0C900)

**SA** Source Address, part of the 29 bit identifier of SAE J1939 messages

**SAE**
Society of Automotive Engineers (in charge of J1939 specification)

**SPN**
Suspect Parameter Number, unique identifier of an SAE J1939 parameter

**XFER**
J1939 Transfer PG (PGN = 0x0CA00)